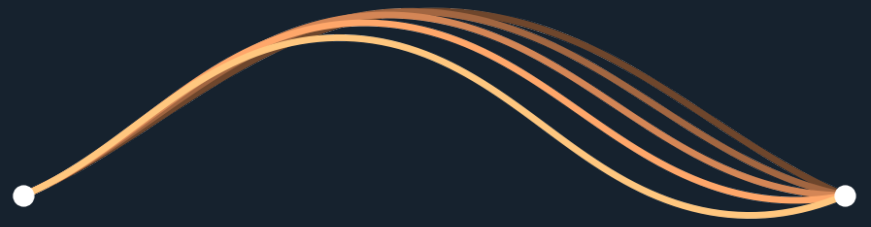
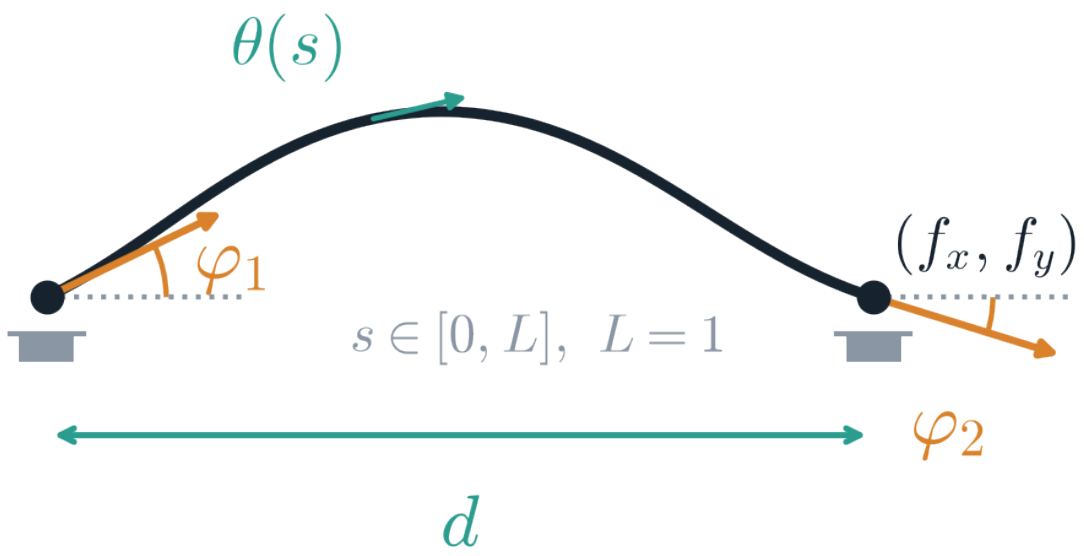
Mapping the clamped elastica across its snap surface

Continuation with buckling-mode branch tracking in FEniCSx / PETSc



One rod, three parameters, a map of equilibria



For every point (d, φ_1, φ_2) solve for the tangent angle θ(s) and the end forces f_x, f_y; record both clamp moments and the bending energy U.

d

End-to-end distance (L = 1)

0.70 – 0.99; Δd = 0.01

φ_1

Clamp angle at s = 0

−π/2 to π/2; Δφ = π/314 ≈ 0.01

φ_2

Clamp angle at s = L

Same range as φ_1; grid 30 × 315 × 315

Equilibria are stationary points of the bending energy

Bending energy

Inextensible Euler–Bernoulli rod, EI = 1

$$\Pi[\theta] = \frac{1}{2}\int_0^L \theta'(s)^2\, ds$$

Constraints

End position fixed, both ends clamped

$$\int_0^L \cos\theta\ ds = d, \quad \int_0^L \sin\theta\ ds = 0, \quad \theta(0) = \varphi_1, \quad \theta(L) = \varphi_2$$

Lagrangian

f_x, f_y enforce the two end-position integrals

$$\mathcal{L}[\theta, f_x, f_y] = \frac{1}{2}\int_0^L \theta'^2\, ds + f_x\left(\int_0^L \cos\theta\ ds - d\right) + f_y \int_0^L \sin\theta\ ds$$

f_x and f_y are the Lagrange multipliers of the end-position constraints — physically, the reaction forces at the right clamp. The study maps every stable stationary point of $\mathscr{L}$ the continuation reaches over the (d, φ_1, φ_2) box.

Stationarity gives the elastica equation and a mixed weak form

Euler–Lagrange equation

$$\theta'' + f_x \sin\theta - f_y \cos\theta = 0$$

$\theta(0) = \varphi_1,\ \theta(L) = \varphi_2$

1. $\theta \in$ P1 Lagrange on 200 elements, graded towards both clamps
2. $f_x, f_y \in$ “Real” space: one global DOF each
3. Clamp angles imposed as Dirichlet BCs on θ

Residuals assembled by FEniCSx (solver.py)

$$F_0 = \int_0^L \left(\theta' v' - f_x \sin\theta\ v + f_y \cos\theta\ v\right) ds = 0 \quad \forall v \in V_0$$

$$F_1 = \int_0^L (\cos\theta - d)\, q_x\, ds = 0$$

$$F_2 = \int_0^L \sin\theta\ q_y\, ds = 0$$

Post-processed per point

$$M_1 = \theta'(0), \quad M_2 = \theta'(L), \quad U = \frac{1}{2}\int_0^L \theta'^2\, ds$$

Moments: end-element slope plus a half-element Taylor correction (second order)

Each parameter point is one Newton solve on 203 unknowns

$$J(\mathbf{u}^k)\,\delta\mathbf{u} = -\mathbf{F}(\mathbf{u}^k), \qquad \mathbf{u}^{k+1} = \mathbf{u}^k + \alpha_k\,\delta\mathbf{u}$$

u = ($\theta_0 \ldots \theta_{200}$, f_x, f_y); ($0 < \alpha_k < 1$) from cubic backtracking line search

203

unknowns per point

201 nodal angles + 2 multipliers

LU

MUMPS direct solve

KSP preonly: exact linear steps

10^{-10}

relative tolerance

absolute 10^{-12}

50

max Newton its

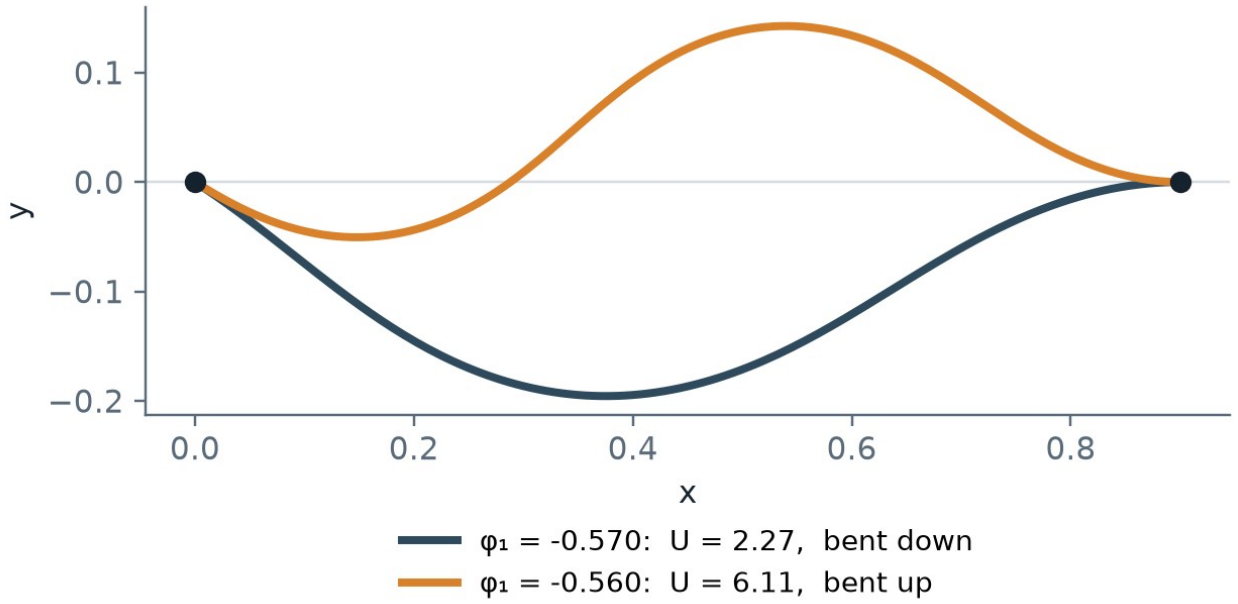
per warm start; more counts as a failure

At a fold of the solution branch, the Jacobian becomes singular, so the snap surface is exactly where the Newton solver fails, and where the continuation logic has to take over.

The snap surface

Where the equilibrium jumps from one branch to another

Two equilibria 0.01 rad apart: the rod has snapped



d = 0.90, φ_2 = 0, production run. Converged solutions on either side of the jump.

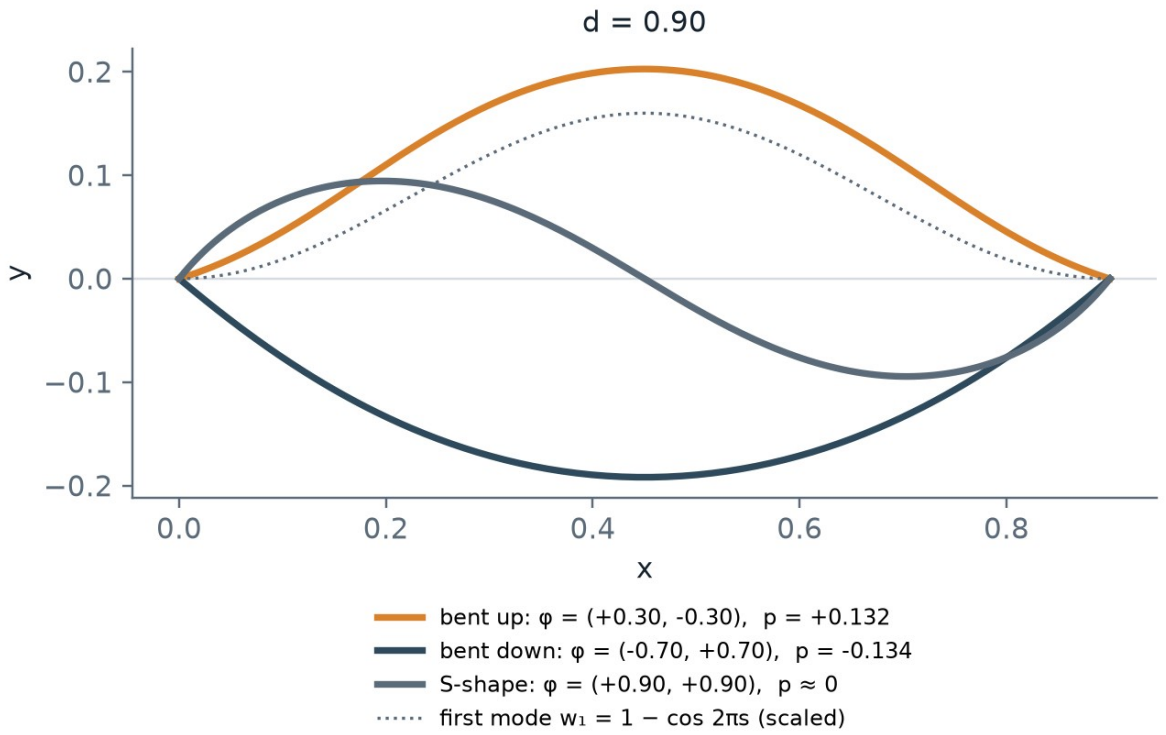
Why Newton struggles here

1. **Fold**
   The branch ends; the Jacobian goes singular and Newton steps blow up.

2. **Silent branch switch**
   A warm start from one side converges, but to the other branch.

3. **Residual can't tell**
   Both sides meet 10^{-10}, so convergence alone says nothing about which branch you are on.

Bent up or bent down: the first buckling mode decides



$$p = \frac{1}{\|w_1\|}\int_0^L y\, w_1\, ds$$

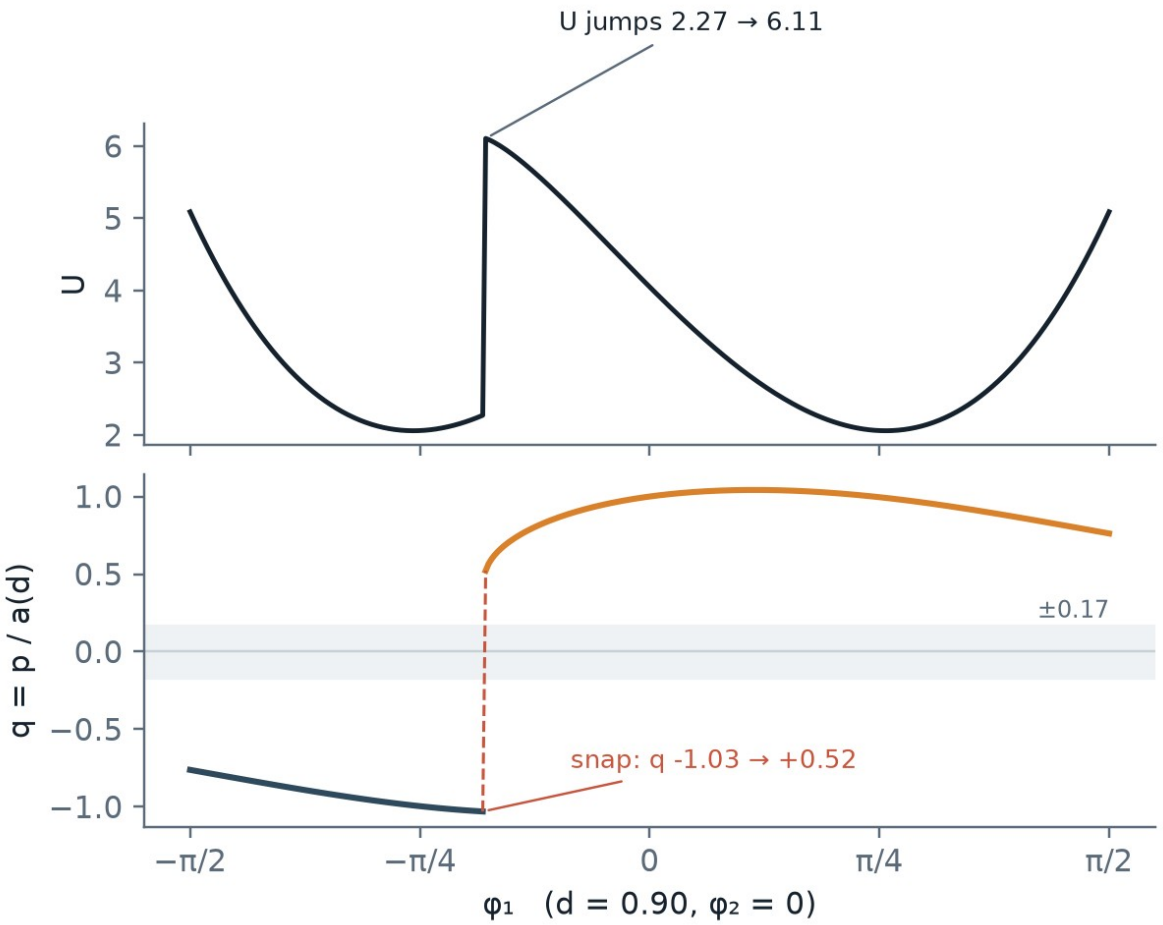
$$y(s) = \int_0^s \sin\theta\, dt, \qquad w_1(s) = 1 - \cos\frac{2\pi s}{L}$$

1 **First clamped buckling mode**
w_1 is the symmetric hump; the second mode is antisymmetric and carries no up/down information.

2 **Sign = branch**
$p > 0$: bent up. $p < 0$: bent down. The S-shape has $p \approx 0$.

3 **Built-in symmetry**
p is unchanged by the mirror $(\varphi_1, \varphi_2) \to (-\varphi_2, -\varphi_1)$ and flips under $(\varphi_1, \varphi_2) \to (-\varphi_1, -\varphi_2)$.

A snap is a sign change of p with a jump



$$q = \frac{p}{a(d)}$$

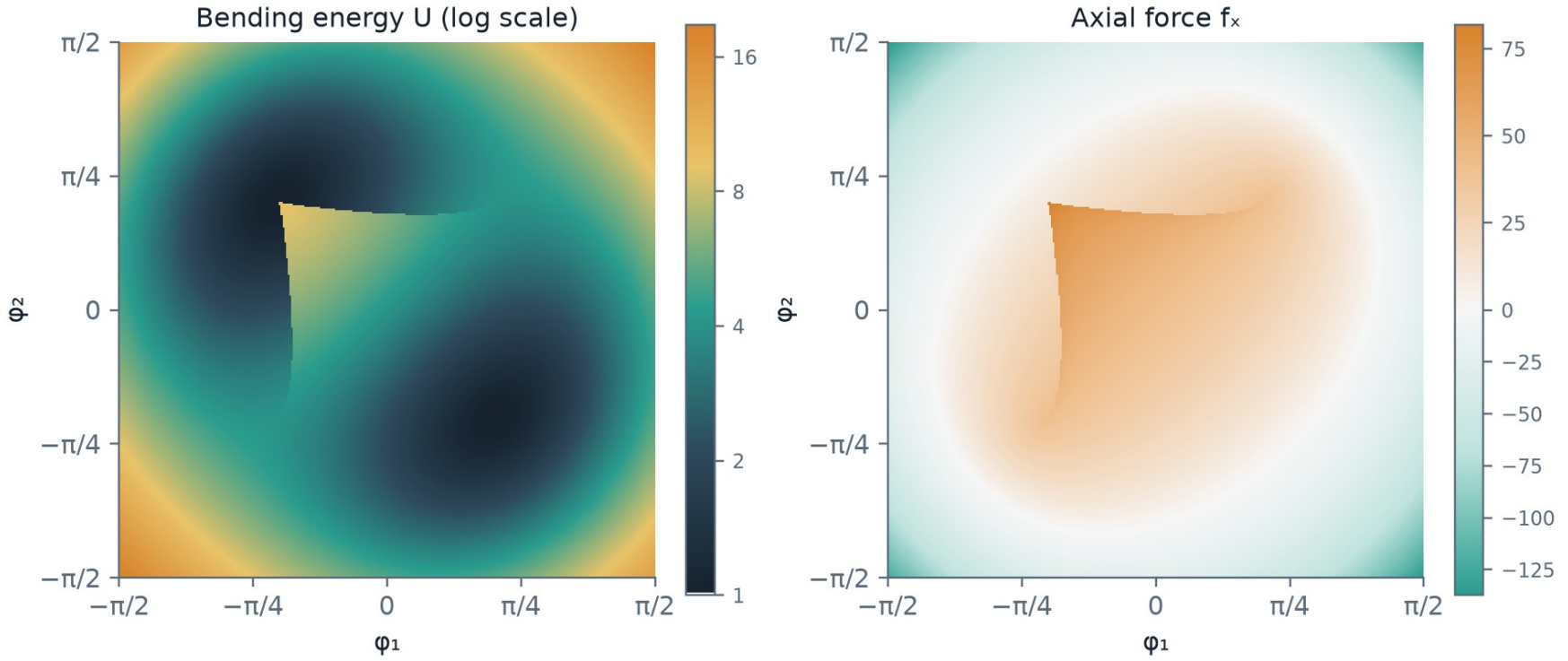
$$|\Delta q| > 0.17 \quad \text{and} \quad |\Delta q| > \frac{1}{2}\max(|q_a|, |q_b|)$$

1. **Scale out d**
   p shrinks like √(1 − d); a(d) is |p| of the exact first mode at (d, 0, 0), so q = ±1 there at every d.

2. **Snap**
   Sign change with a jump: q goes −1.03 → +0.52 in one 0.01 step. Smooth neighbours differ by about 0.003.

3. **Smooth crossing**
   Sign change without a jump, through the S-shape near $\varphi_1 = \varphi_2$. Not a snap.

Smooth fields, then a jump across an L-shaped surface



Production run, $d = 0.90$ layer: $\varphi_1, \varphi_2 \in [-\pi/2, \pi/2]$, $\Delta\varphi \approx 0.01$, 99,225 points.

Smooth interior

Grid neighbours differ by $|\Delta q| \approx 0.003$

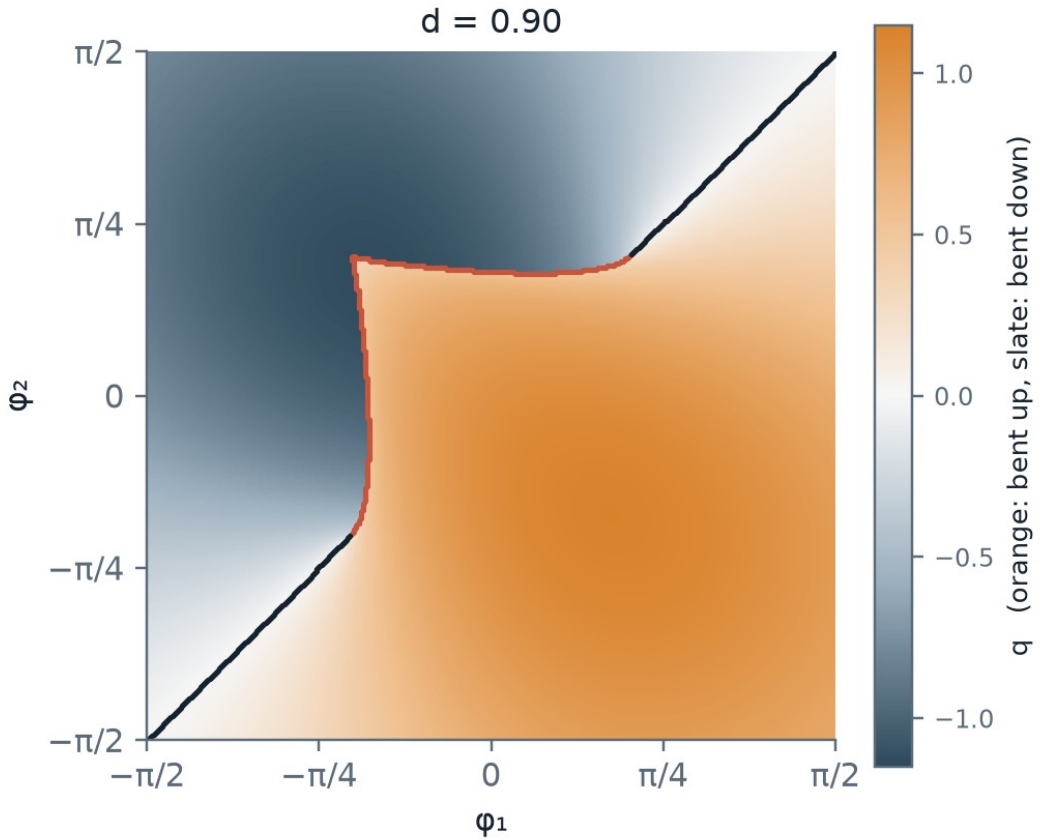
Sharp edge

Jump at $\varphi_1 \approx -0.57$ and, by symmetry, $\varphi_2 \approx +0.57$

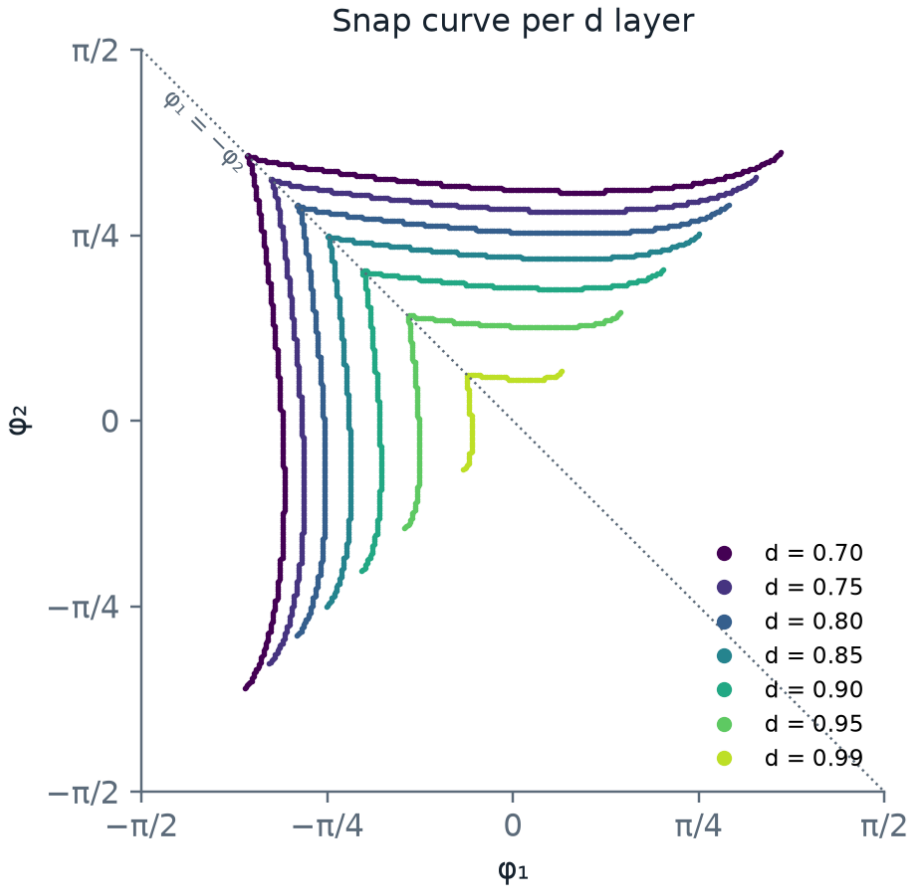
Large jump

U 2.27 → 6.11 and f_x 28 → 59 in a single 0.01 step

The sign of p traces the snap surface

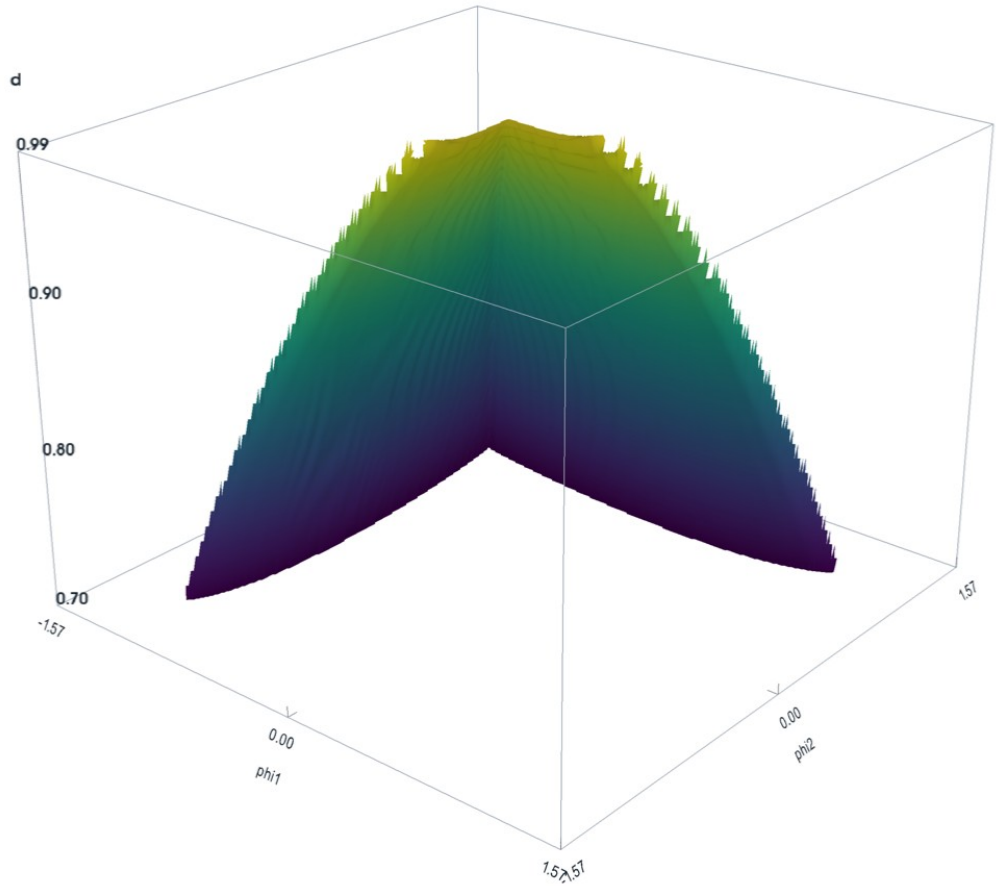


Red: snap edges, where p changes sign with a jump. Black: smooth crossings through the S-shape.

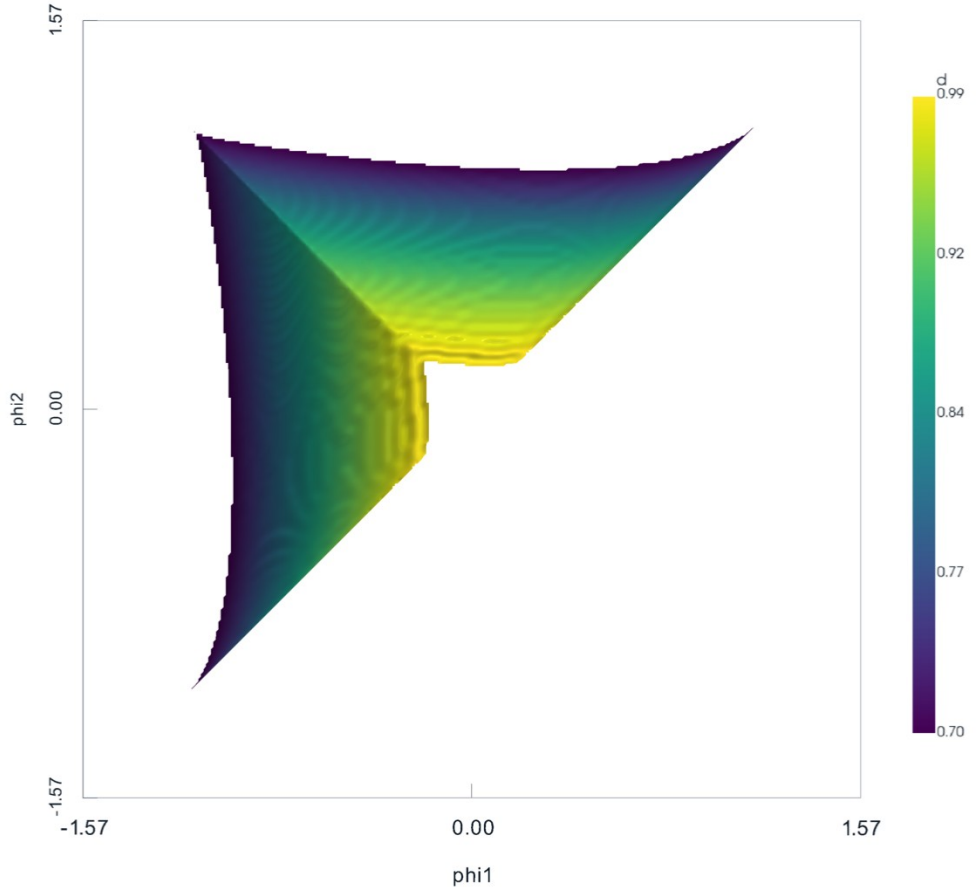


As d → 1 the L closes in: at $\varphi_2 = 0$ the snap sits at $\varphi_1 \approx -0.98$ (d = 0.70), −0.57 (0.90) and −0.18 (0.99).

The snap surface in 3D

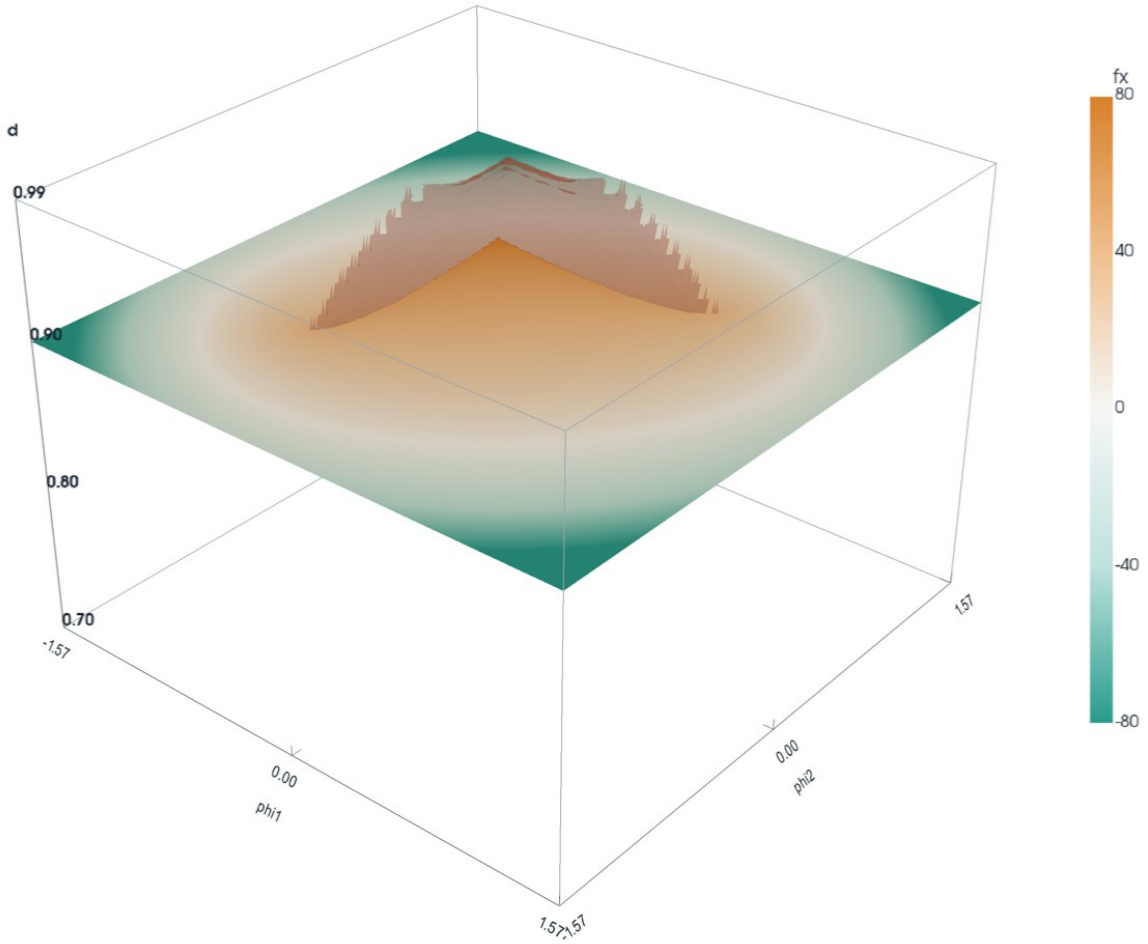


Oblique view, d stretched ×8. The p = 0 contour of the 17.2 M-tetrahedron grid, kept where p jumps: 30,081 snap edges, area 2.64.



Down the d axis: every layer is the L-shape, and the sheet closes in on the origin as d → 1. Colour: d.

The sheet rises from the step in axial force



Layer d = 0.90 coloured by f_x (clipped at ±80); the snap sheet above it is translucent.

Across the snap at $\varphi_2 = 0$, f_x roughly doubles at every d:

30.1 → 61.5 — d = 0.80, jump at $\varphi_1 \approx -0.80$

28.3 → 58.7 — d = 0.90, jump at $\varphi_1 \approx -0.57$

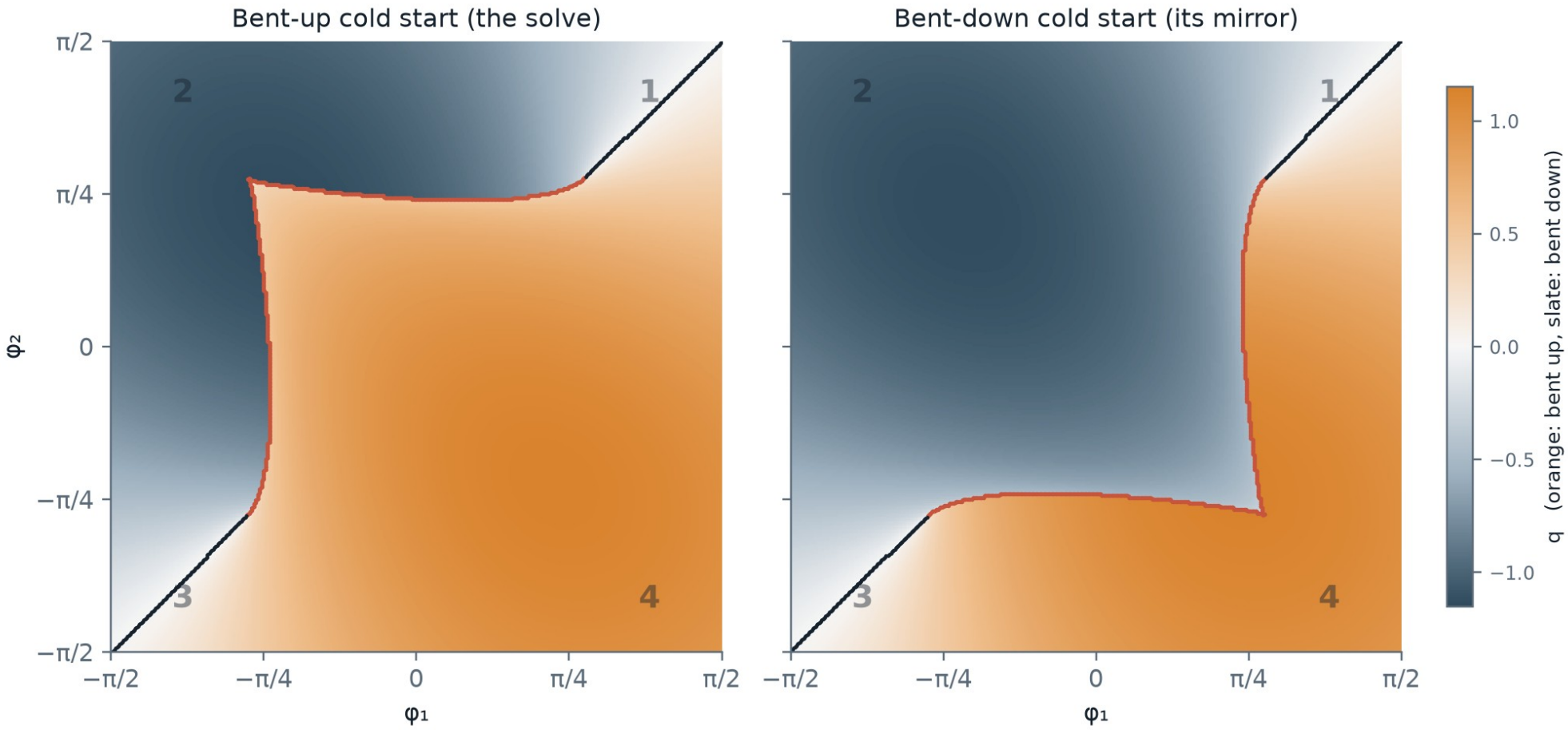
27.4 → 55.4 — d = 0.97, jump at $\varphi_1 \approx -0.31$

The surface is built from the shape alone, through the sign of p, yet it lands exactly on the step the forces see.

The bistable region

Two stable rods for the same clamps, and which snap the rod can meet

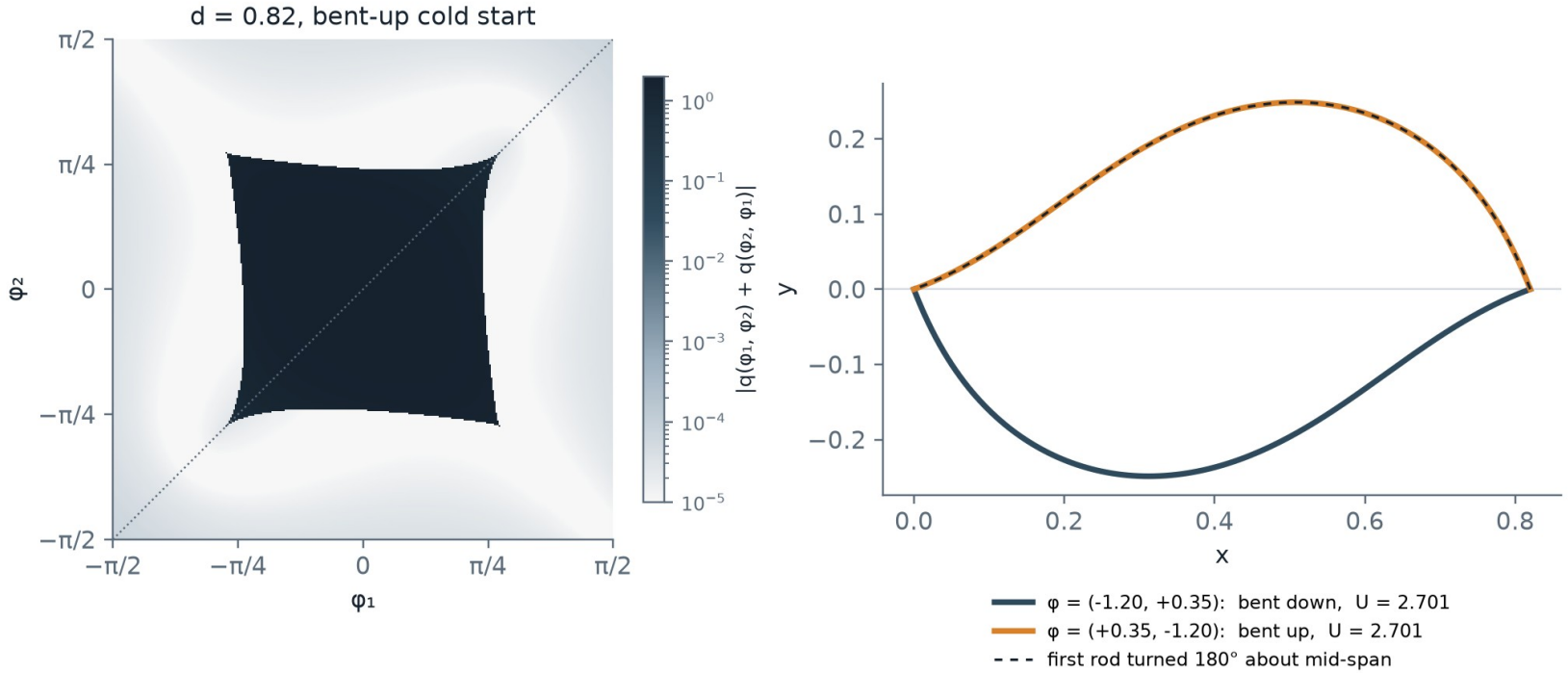
A bent-down cold start mirrors the map across $\varphi_1 = \varphi_2$



d = 0.82. θ → −θ takes the solve to the bent-down cold start: $(\varphi_1, \varphi_2) \to (-\varphi_1, -\varphi_2)$ with p negated. With the mirror $(\varphi_1, \varphi_2) \to (-\varphi_2, -\varphi_1)$ that is a reflection in $\varphi_1 = \varphi_2$ with orange and slate swapped. Red: each map's snap curve, in quadrants 1, 2, 3 and 1, 3, 4.

Outside the region, swapping the clamps turns the rod over

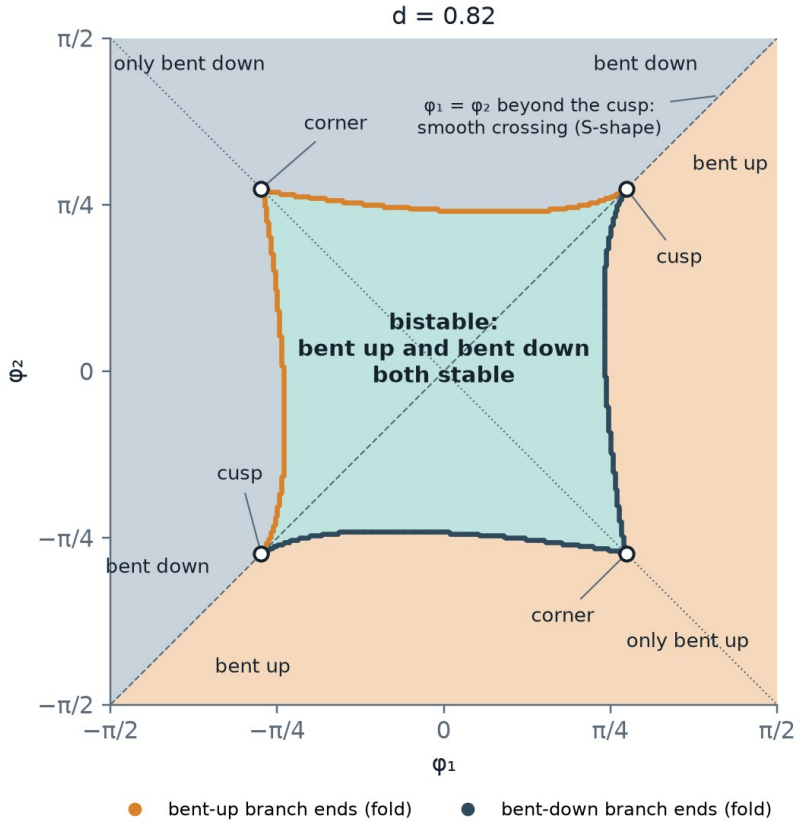
$$(d, \varphi_1, \varphi_2) \mapsto (d, \varphi_2, \varphi_1): \quad \tilde{\theta}(s) = \theta(L - s), \quad \tilde{p} = -p, \quad \tilde{U} = U, \quad \tilde{f}_x = f_x, \quad \tilde{f}_y = f_y, \quad \tilde{M}_1 = -M_2$$

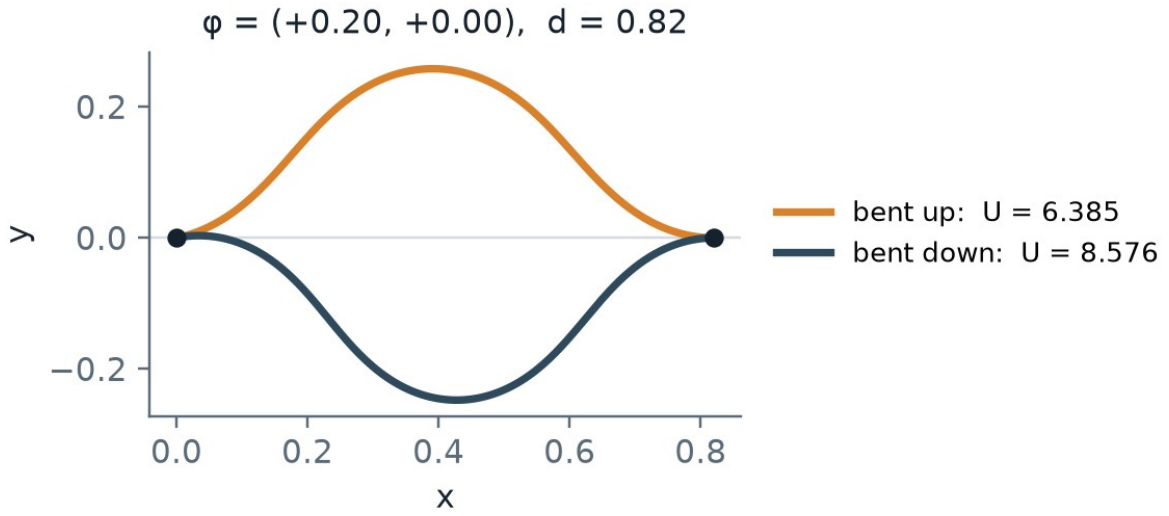


Left: $|q(\varphi_1, \varphi_2) + q(\varphi_2, \varphi_1)|$ at d = 0.82 is about 1e-4 everywhere except a square, the bistable region. Over all layers 2.35 M of the 2.98 M nodes pass, with q within 6.6e-4 and U within 1.5e-8. Right: the rod at (−1.20, +0.35) turned 180° lies on the rod at (+0.35, −1.20).

The bistable region: two stable rods for the same clamps

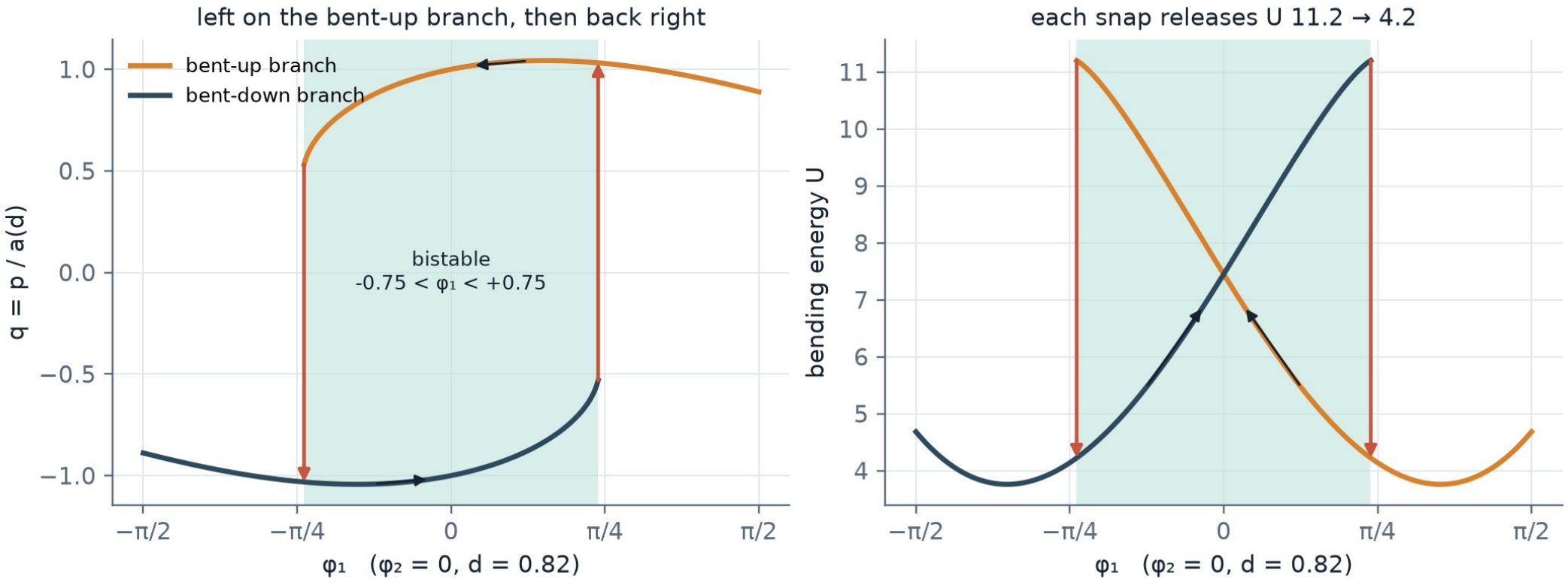
Bistable where the bent-up solve is bent up at a point and at its mirror: $q(\varphi_1, \varphi_2) > 0$ and $q(\varphi_2, \varphi_1) > 0$





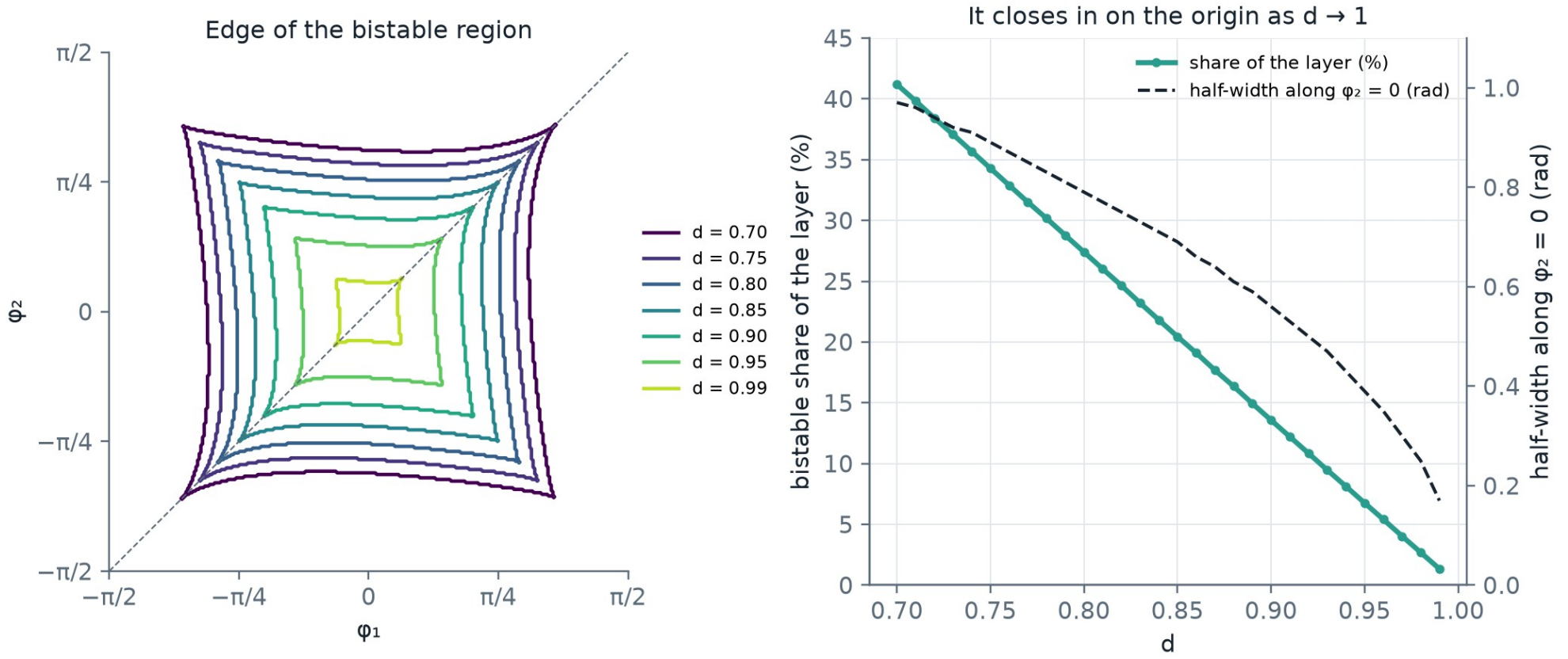
1. Left and top: the bent-up branch ends (fold). Leave this way bent up and the rod snaps down.
2. Bottom and right: the bent-down branch ends, the mirror image. Leave this way bent down and the rod snaps up.
3. On the diagonal the curves meet at cusps. Beyond them $\varphi_1 = \varphi_2$ is a smooth S-crossing and nothing snaps.

Which snap the rod meets depends on the branch it is on



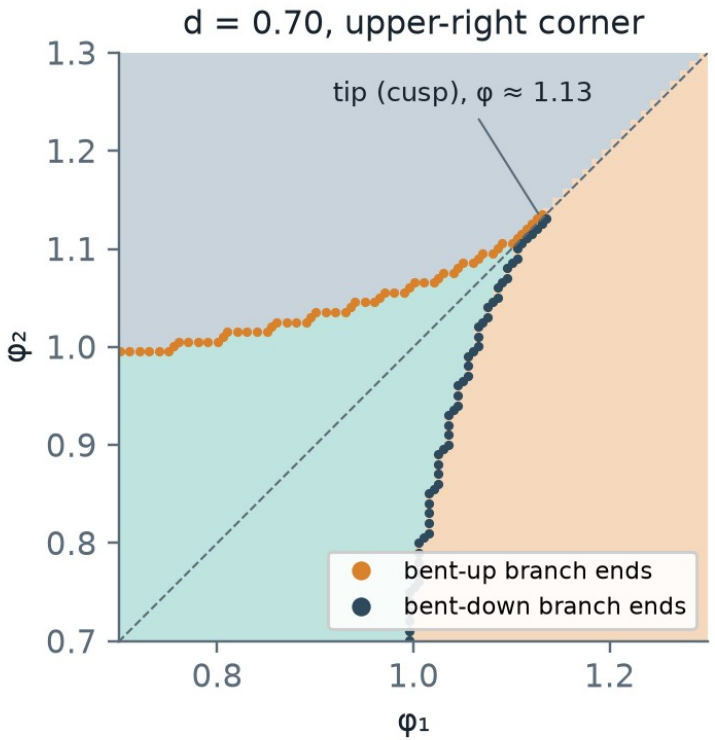
d = 0.82, $\varphi_2 = 0$. Bent up (orange), the rod crosses the whole bistable band (teal) and snaps down only at $\varphi_1 \approx -0.76$. Bent down (slate), it snaps up only at +0.76. The loop between the two snaps is the hysteresis; each snap releases U 11.2 → 4.2.

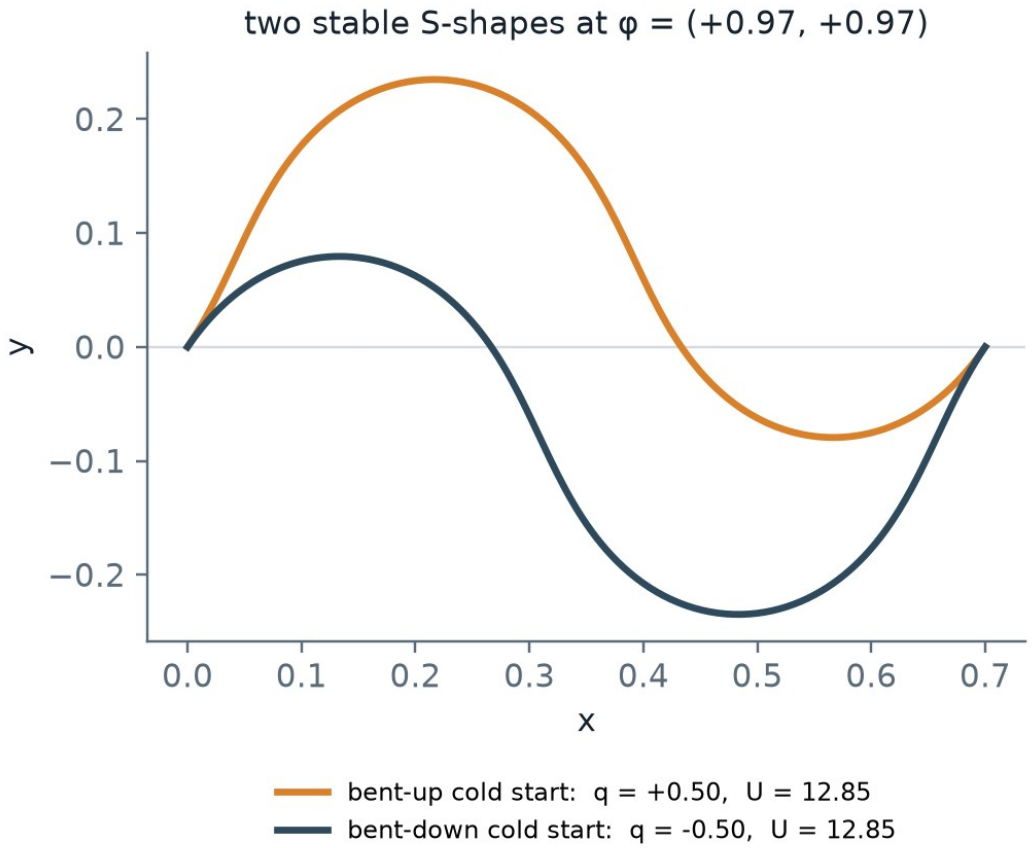
The bistable region closes in as d → 1



Share of the layer: 41 % at d = 0.70, 25 % at 0.82, 14 % at 0.90, 1.3 % at 0.99. Along $\varphi_2 = 0$ it spans $|\varphi_1| < 0.97$, 0.75, 0.56 and 0.17. The largest energy gap between the two rods falls from 25.6 to 0.8.

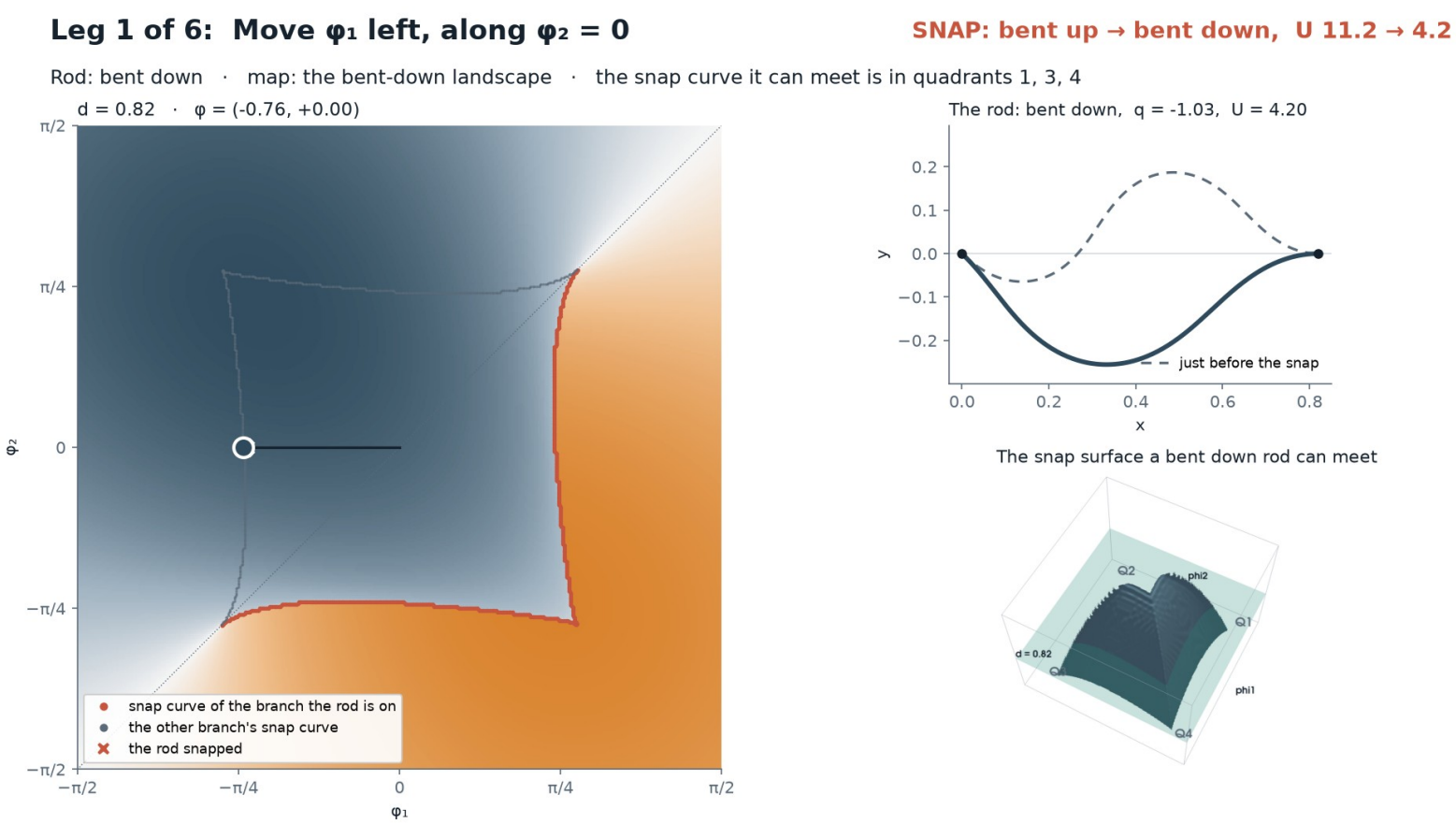
The region ends in a cusp on the diagonal





d = 0.70. Towards the diagonal the two fold curves close in and meet at a cusp at φ ≈ 1.13. Near it both stable rods are S-shapes, mirror images with q ≈ ±0.5 and equal energy on the diagonal; at the cusp q falls to 0. The cusp moves in with d: φ ≈ 0.86 at d = 0.82, 0.20 at 0.99.

Following one rod around the snap surface



Click to play (29 s). d = 0.82. Left: the map of the branch the rod is on; red, the snap curve it can meet; grey, the other branch's. Right: the rod, and the snap surface in (φ₁, φ₂, d) for its branch.

1. Across. Bent up, the rod snaps down at $\varphi_1 = -0.76$, and the map and snap surface switch to the bent-down ones. Coming back it snaps up only at +0.76.

2. Around. Raising φ_2 at $\varphi_1 = 1.25$ passes the end of the snap curve: a smooth S-crossing turns the rod bent down, no snap.

3. Back at the start, now bent down. Repeating the first move, nothing snaps: its snap curve is in quadrants 1, 3, 4.

Continuation strategy

How the solver walks the parameter space and guards against branch switches

Traverse the grid; reject jumps and unstable results

Cold-start once at $\varphi_1 = \varphi_2 = 0$, smallest d; visit nodes by grid distance from it

→

Warm-start θ, f_x, f_y from a solved grid neighbour: bent-up ones first, then nearest

→

Newton solve (SNES, at most 50 iterations)

→

No jump in q, and stable?

yes →

Accept, store, and write the mirror point

no ↓

Rejected or diverged: try the next neighbour. After 3 failures from one branch, wait for the other; a waiting node retries whenever a neighbour is solved.

Grid

Rectilinear (d, φ_1, φ_2) grid split into 6 tetrahedra per cell; warm starts come from the 26 grid neighbours

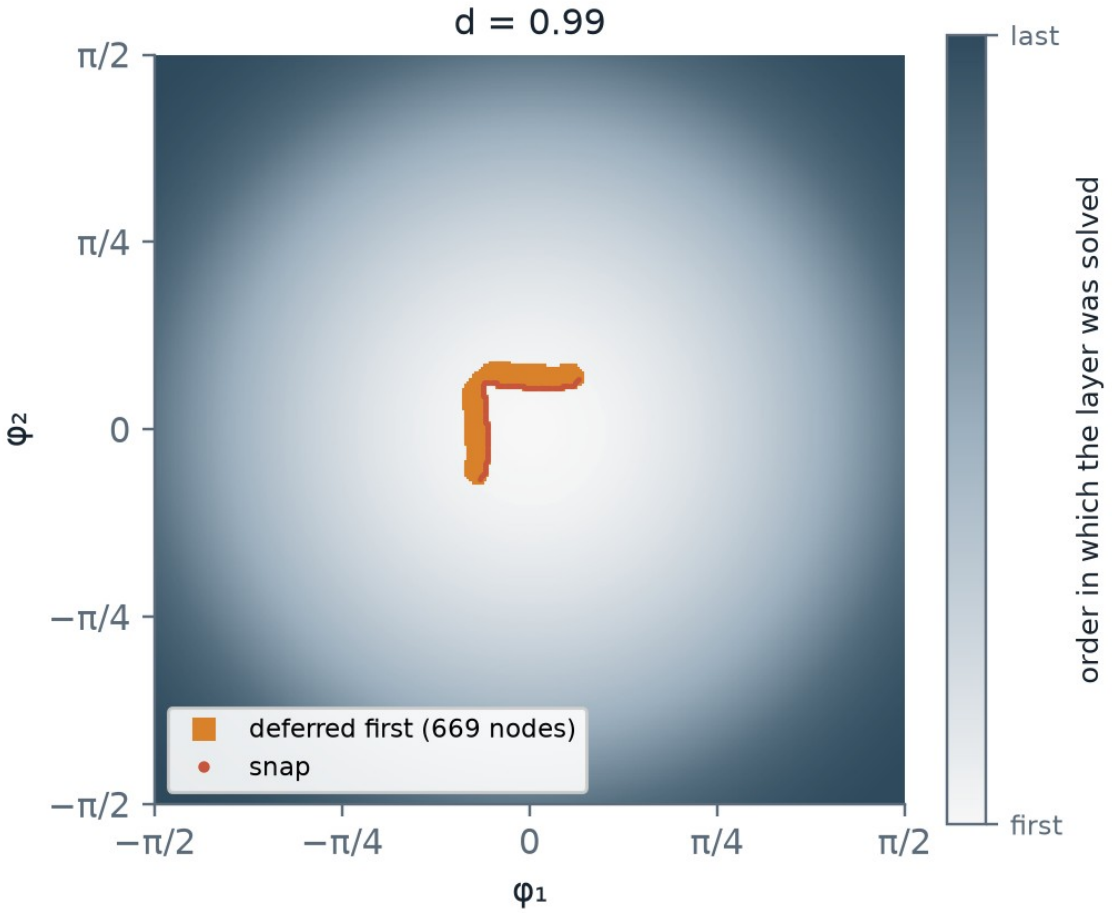
Restartable

Live NPZ checkpoints, also on Ctrl-C; a resume reloads every stored θ as a warm-start source

Half domain

Only $\varphi_1 + \varphi_2 \geq 0$ is solved; a precomputed mirror index fills the other half

Going around a fold: defer, never jump



d = 0.99 layer: the front grows outward; orange nodes had to wait for a neighbour on the other branch.

1. **Defer**
   A node with no acceptable warm start is set aside, not forced; it is retried each time a new neighbour is solved.

2. **Wait for the other branch**
   After 3 failures from neighbours on one branch it only tries the other, so a node past a fold waits for the far side.

3. **The long way round**
   The front reaches the cut-off region through the smooth crossings and approaches it from the far side.

4. **Last resort**
   Only if the whole front stalls is the nearest waiting node cold-started.

Production run: 1,155 nodes waited (all at d ≥ 0.97); one extra cold start, at (0.99, 0.20, 0.20). A final promotion pass gave the bent-up rod to 1,571 bistable nodes the front had filled bent down.

A stability test catches what the jump test misses

Constrained second variation

$$Q[v] = \int_0^L v'^2 - (f_x \cos\theta + f_y \sin\theta)\, v^2 \, ds$$

$$v(0) = v(L) = 0, \qquad \int_0^L v\cos\theta \, ds = \int_0^L v\sin\theta \, ds = 0$$

$$n_- = \mathrm{neg}(A) + \mathrm{pos}(C^T A^{-1} C) - 2$$

Stable when $n_- = 0$: no direction lowers the energy.

17

unstable results rejected

during the production run

0

unstable in the final map

all 2,976,750 nodes re-tested

1

Why the jump test is not enough

On $\varphi_1 = -\varphi_2$ the rod is symmetric ($f_y = 0$); the bent-up branch loses stability at a symmetry-breaking bifurcation, not a fold. Newton follows it smoothly, with no jump in q.

2

Inertia of a bordered matrix

A is the tridiagonal P1 form, C the two constraint vectors (Haynsworth). One LDL^T sweep: O(N), about 0.2 ms against an 11 ms solve.

3

Every candidate

Warm starts and cold starts alike: an unstable result counts as a failure and the node moves on to the next neighbour.

Deriving the exact first mode used for the cold start

1. $\varphi_1 = \varphi_2 = 0$ is mirror-symmetric, so $f_y = 0$ and the elastica is a pendulum

$$\theta'' + \lambda^2 \sin\theta = 0, \quad \lambda^2 = f_x, \quad f_y = 0$$

2. Multiply by θ' and integrate once; $\theta' = 0$ at the peak angle α

$$\theta'^2 = 4\lambda^2\left(k^2 - \sin^2\tfrac{\theta}{2}\right), \qquad k = \sin\tfrac{\alpha}{2}$$

3. Substituting $\sin(\theta/2) = k \sin\psi$ gives $\psi' = \lambda\sqrt{(1 - m \sin^2\psi)}$, so $\sin\psi = \mathrm{sn}(\lambda s \mid m)$

$$\theta(s) = 2\sin^{-1}[k\,\mathrm{sn}(\lambda s \mid m)], \qquad m = k^2$$

4. $\theta(L) = 0$ and $\int \sin\theta\, ds = 0$ need one full period of sn (4K): four quarter-waves

$$\lambda L = 4K(m) \quad \Rightarrow \quad f_x = \left(\frac{4K(m)}{L}\right)^2$$

5. The end distance fixes m; d/L falls monotonically in m, so a 1-D root-find is safe

$$\frac{d}{L} = \frac{1}{L}\int_0^L \cos\theta\; ds = \frac{2E(m)}{K(m)} - 1$$

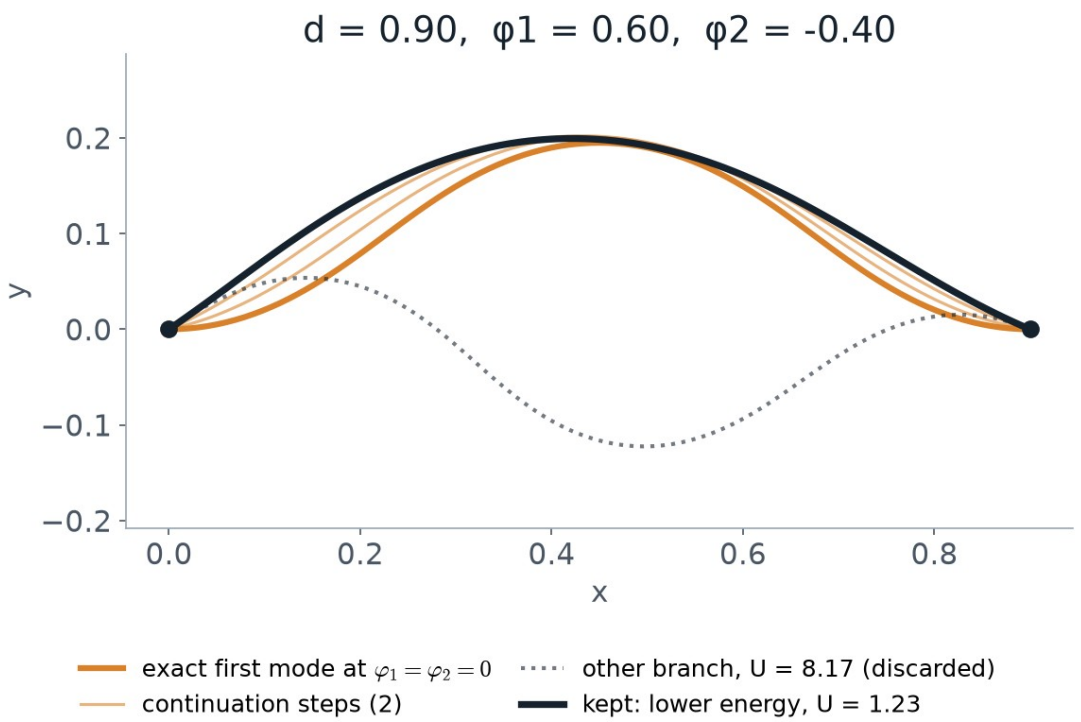
6. Small bow, $m \to 0$: the $A \sin(2\pi s/L)$ shape with $A = 2\sqrt{(1 - d/L)}$, and the Euler load

$$\theta \approx 2\sqrt{m}\,\sin\frac{2\pi s}{L}, \quad m \approx 1 - \frac{d}{L}, \quad f_x \to \frac{4\pi^2}{L^2}$$

$$K(m) = \int_0^{\pi/2} \frac{d\psi}{\sqrt{1 - m\sin^2\psi}}, \qquad E(m) = \int_0^{\pi/2} \sqrt{1 - m\sin^2\psi}\; d\psi$$

Exact first-mode cold start, then continue the clamp angles

$$\theta \approx 2\sqrt{m}\,\sin\frac{2\pi s}{L}, \quad m \approx 1-\frac{d}{L}, \quad f_x \to \frac{4\pi^2}{L^2}$$



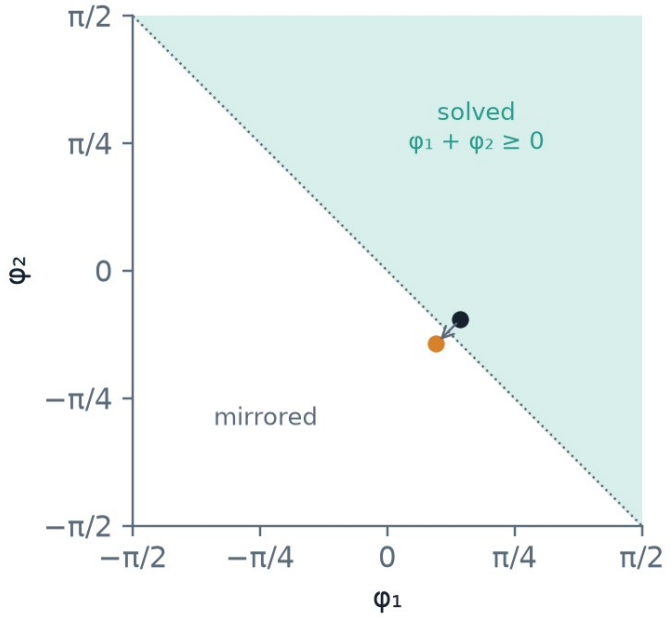
Predictor for each continuation step

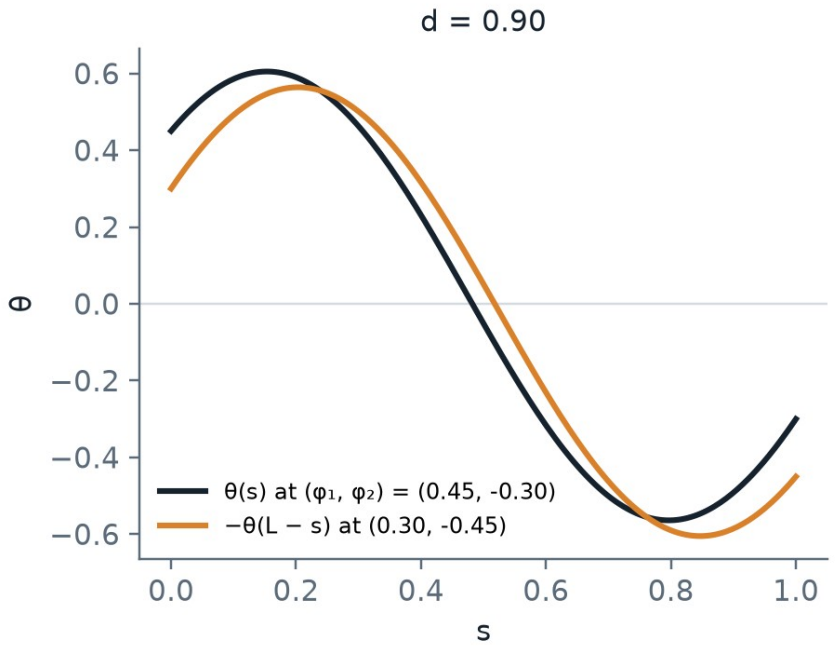
$$\theta_0^{(k+1)}(s) = \theta^{(k)}(s) + \Delta\varphi_1\left(1-\frac{s}{L}\right) + \Delta\varphi_2\,\frac{s}{L}$$

1. Exact at $\varphi_1 = \varphi_2 = 0$ for any d, with $\int \sin\theta\, ds = 0$: Newton converges in 2 iterations
2. Steps ≤ 0.25 rad, ≤ 15 Newton iterations each; a failed step halves, giving up below 0.02 rad (a fold)
3. Both hump signs are continued and the lower energy kept, so the result does not depend on which branch the ramp follows

Mirror symmetry halves the number of solves

$$(d, \varphi_1, \varphi_2) \mapsto (d, -\varphi_2, -\varphi_1): \quad \tilde{\theta}(s) = -\theta(L-s), \quad \tilde{f}_x = f_x, \quad \tilde{f}_y = -f_y, \quad \tilde{M}_1 = M_2, \quad \tilde{p} = p$$





1. Reflecting the rod about mid-span maps (φ_1, φ_2) to $(-\varphi_2, -\varphi_1)$
2. Only $\varphi_1 + \varphi_2 \geq 0$ is solved; points on the anti-diagonal are solved once
3. f_x, U and p carry over, f_y flips sign, and the left and right moments swap

Final run

$d \in [0.70, 0.99]$, $\varphi_1, \varphi_2 \in [-\pi/2, \pi/2]$, $\Delta = 0.01$: 2.98 M equilibria and their level sets

2.98 M equilibria, every one checked

2,976,750

equilibria

30 × 315 × 315 grid; 1.49 M Newton solves, the mirror fills the rest

≈ 4 h

on one core

about 11 ms per solve, from one cold start (plus one more near d = 1)

0.27 %

max error vs. BVP

300 random nodes re-solved with scipy solve_bvp; median 0.014 %

0

unstable equilibria

stability test on all 2,976,750 nodes

Exact checks pass

Mirror pairs agree to 3×10^{-8}. At $\varphi_1 = \varphi_2 = 0$ every layer matches the exact first mode: f_x within 1.3×10^{-4}.

Branch share

Bent up on 70.6 % of the layer at d = 0.70, falling almost linearly to 50.5 % at d = 0.99. Snap surface area 2.64 (30,081 snap edges).

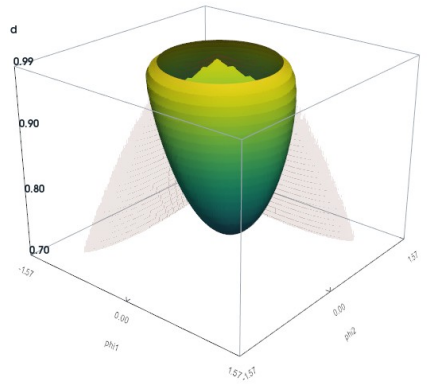
Ranges

f_x from −13,785 (tension, d = 0.99, $|\varphi| = \pi/2$) to +93; $|f_y| \le 340$; $|M| \le 168$; U from 0.10 to 142.

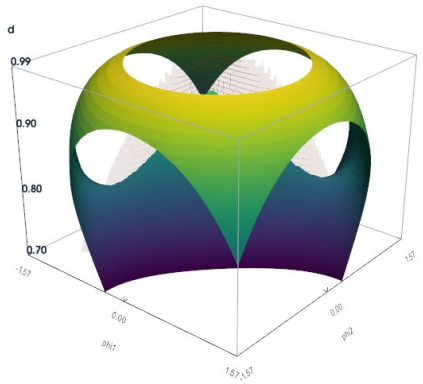
ParaView: data/production/post/elastica_solution_space.vtu (every node, all fields).

Constant-energy surfaces: shells around the straight rod

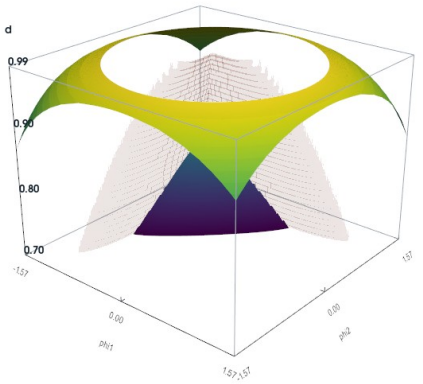
U = 2



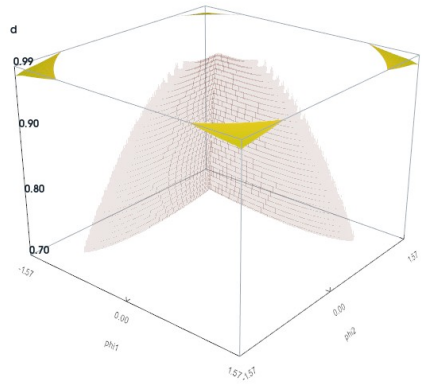
U = 8



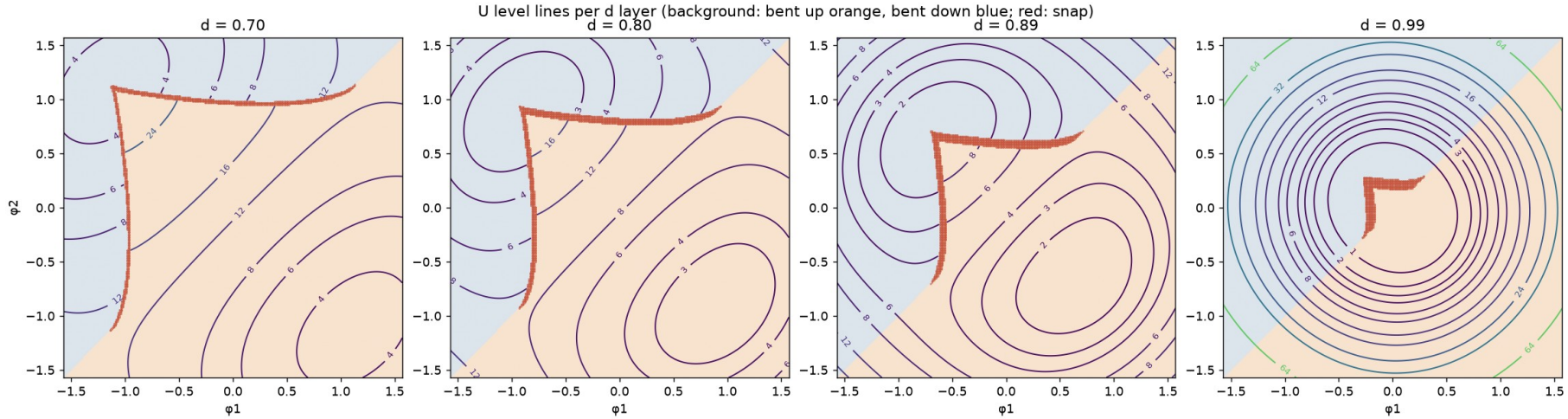
U = 16



U = 64



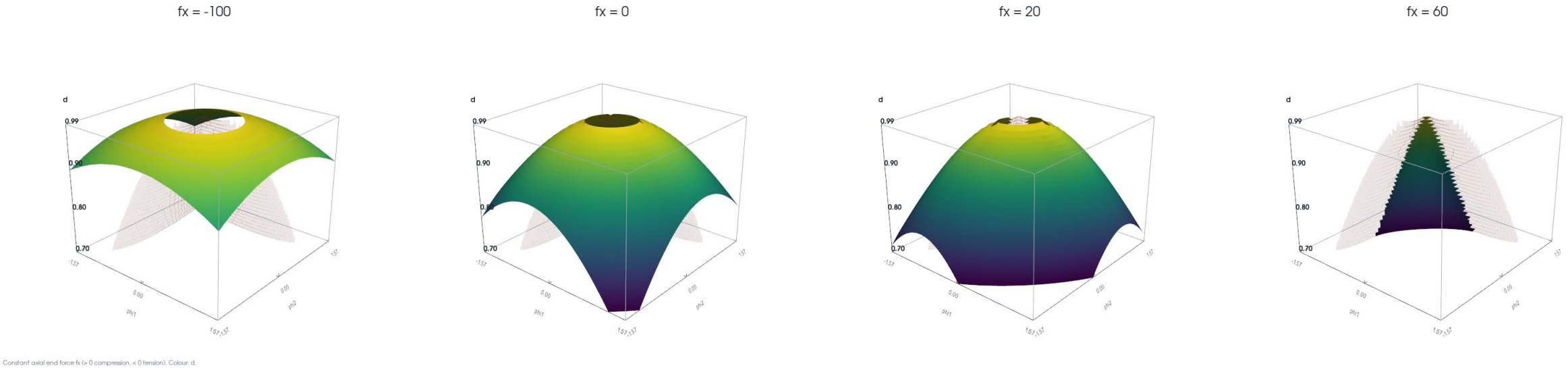
Constant bending energy U = 1/2 ∫ theta'^2 ds (B = L = 1). Colour: d. Faint red: snap part of the p = 0 surface; level sets stop at it.

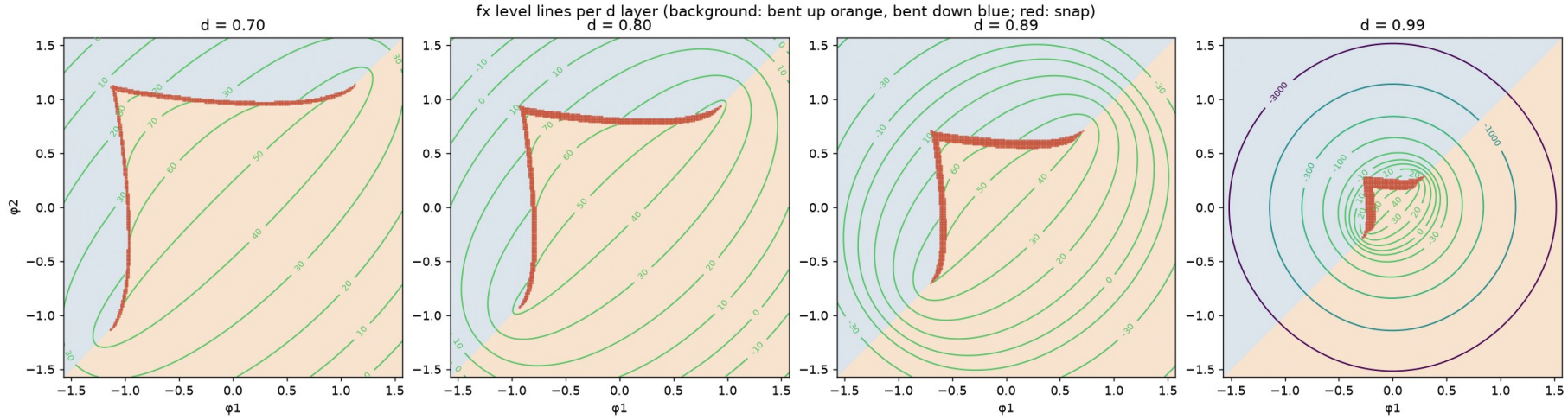


$U = \frac{1}{2}\int \theta'^2\, ds$

From 0.10 to 142. Lowest for a nearly straight rod ($d \to 1$, $\varphi \to 0$): U = 2 is a bowl around that corner and higher levels are wider domes. Each ends at the snap (faint red), where U jumps.

Constant axial force f_x: bounded in compression

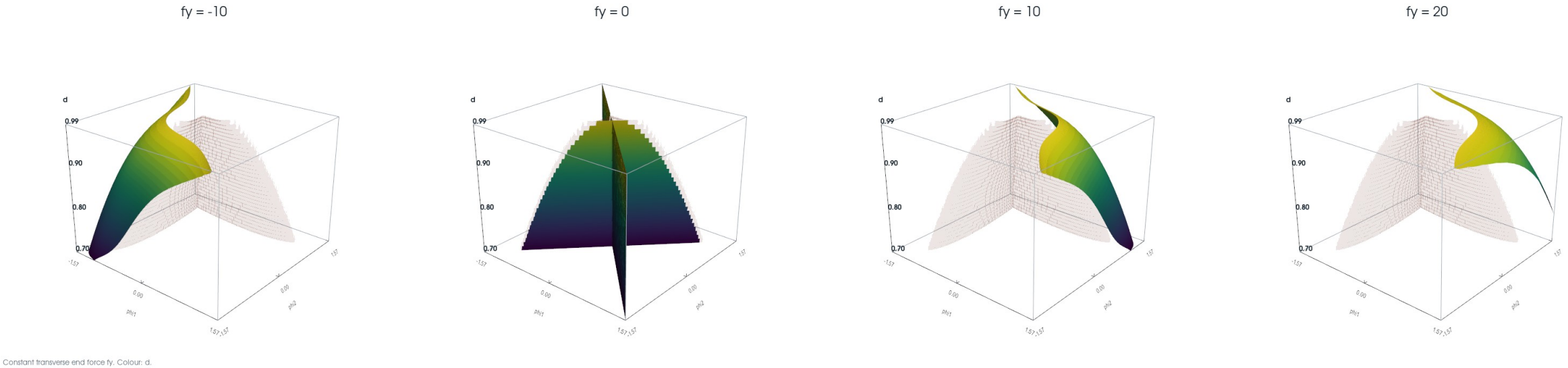




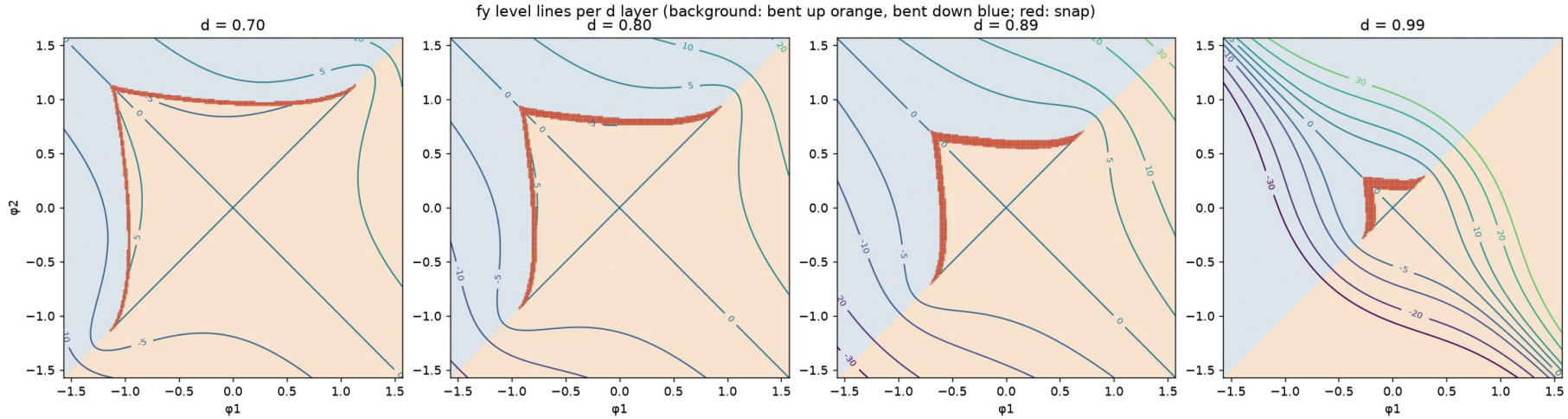
$f_x > 0$: compression

From −13,785 (tension, d = 0.99, $|\varphi| = \pi/2$) to +93. Compressive levels exist only inside the bent-up region: $f_x = 60$ is a sliver and nothing reaches 100.

Constant transverse force f_y: mirror-image pairs



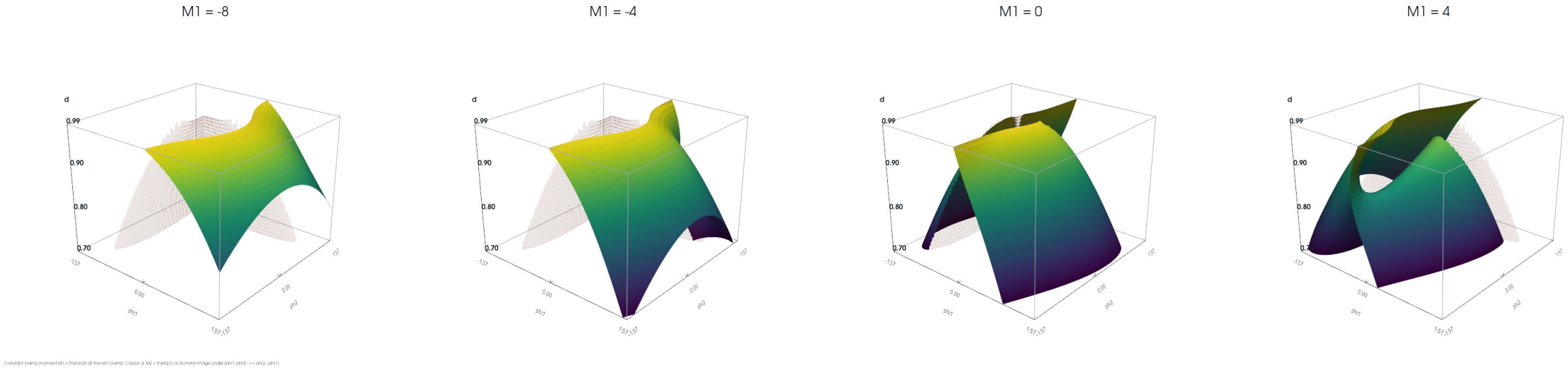
Constant transverse end force fy. Colour: d.

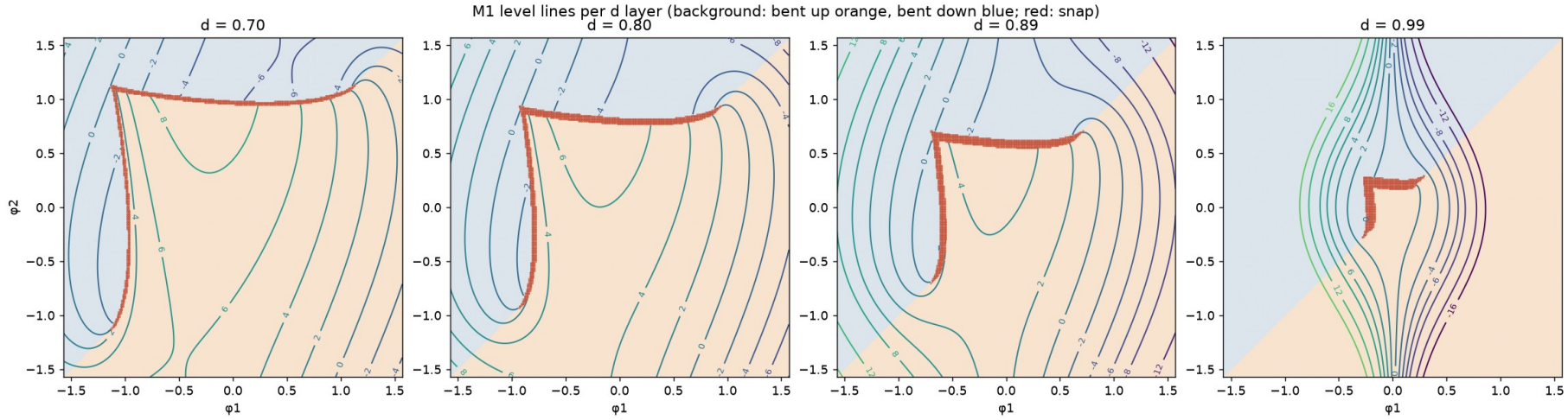


Odd under the mirror

$|f_y| \le 340$. $(\varphi_1, \varphi_2) \to (-\varphi_2, -\varphi_1)$ flips f_y, so $f_y = 0$ holds on the whole plane $\varphi_1 = -\varphi_2$ and ± levels come in pairs of equal area.

Constant clamp moment M_1; M_2 is its mirror image





$M_1 = \theta'(0)$

Within ±168. M_2 at (φ_1, φ_2) is M_1 at $(-\varphi_2, -\varphi_1)$, so its surfaces are mirror images (both in the VTU). Small $|M_1|$ levels split into two sheets.

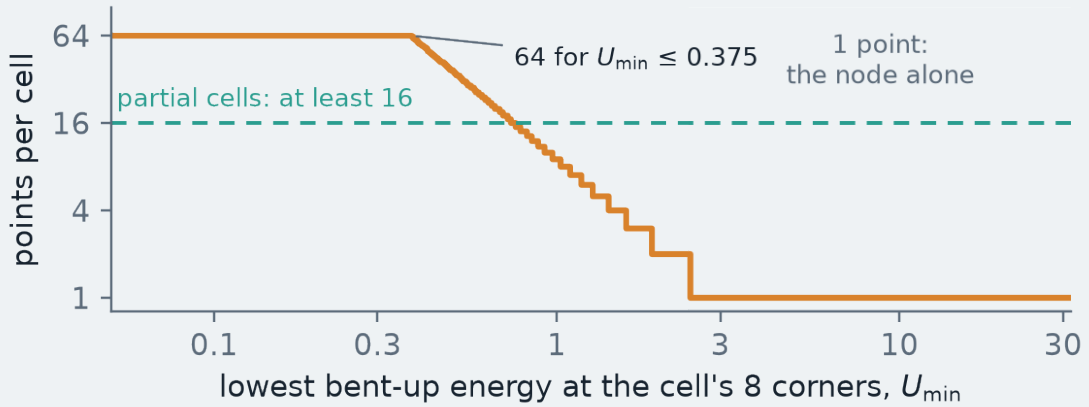
Refined sampling

Training data: both branches at every node, and more solves where the energy is low: 7.6 M states

More points per cell where the energy is low

Points per grid cell

$$n = \mathrm{clip}\left(\left(\frac{u_{\mathrm{ref}}}{U_{\min}}\right)^2,\ 1,\ 64\right)$$



1.11 M

new points solved

998 k states, 112 k past a fold; 23 min on 14 workers

7.6 M

states in the dataset

28.9 % with U < 1, against 2.4 % in the production run

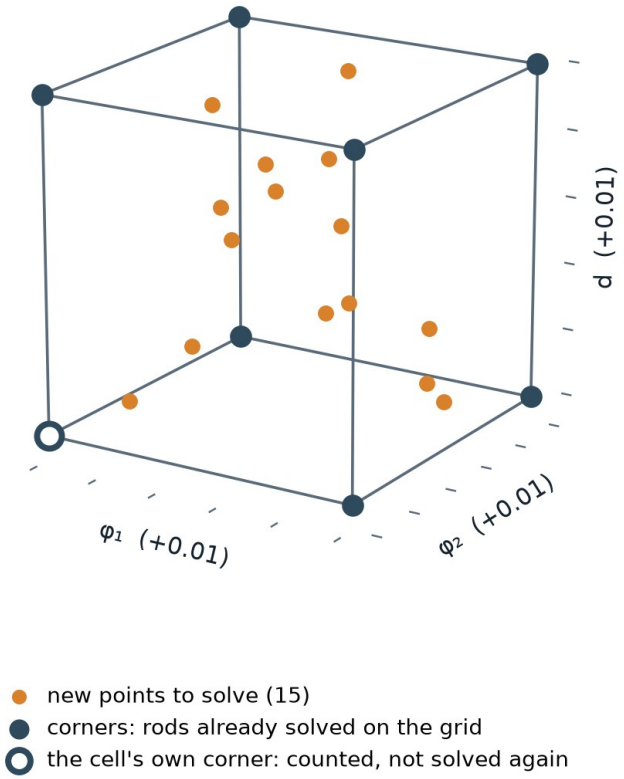
1. **Density follows the energy**
   U_{min} is the lowest bent-up energy at the cell's 8 corners and u_{ref} = 3: one point (the node) above U_{min} = 2.45, all 64 below 0.375.

2. **Partial cells: at least 16**
   Cells where the bent-up branch holds at some corners only: its fold edges, and along $\varphi_1 = \varphi_2$, where p changes sign smoothly.

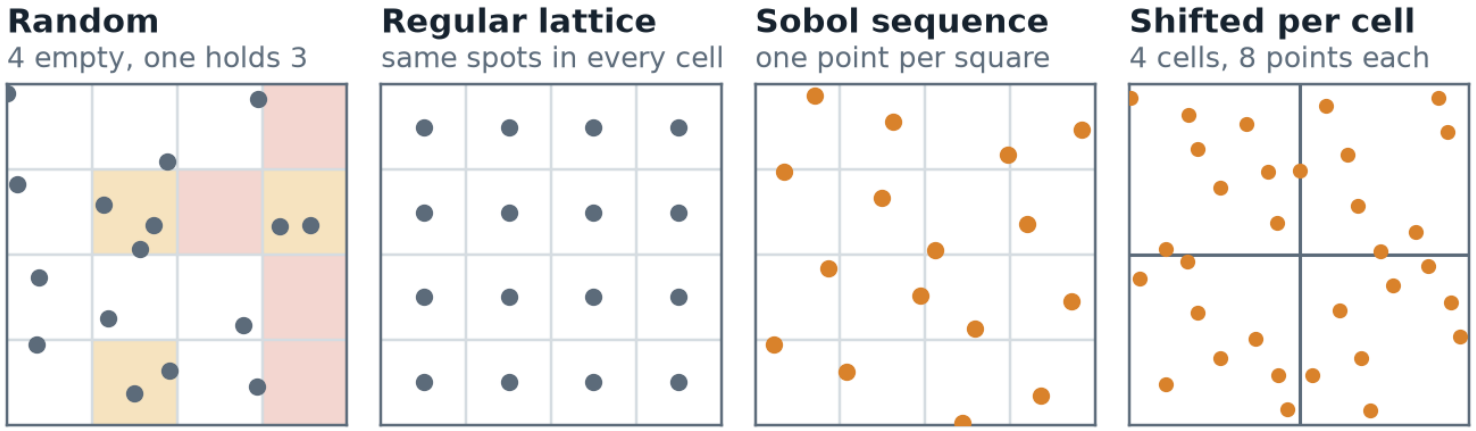
3. **Spread through the cell**
   The node, then a scrambled Sobol set with a random shift per cell. Only the bent-up branch on $\varphi_1 + \varphi_2 \ge 0$ is solved; mirror and reflection give the rest.

4. **Production tests, and negatives**
   Warm start from the corners, nearest first, at most 3 tries: no jump in q, stable. A failure in a partial cell is kept as a negative example (no state).

Inside a cell: new points spread evenly



Example: n = 16, so the cell's own corner plus 15 new points, anywhere inside, d included: the data also covers values between the 30 d layers.



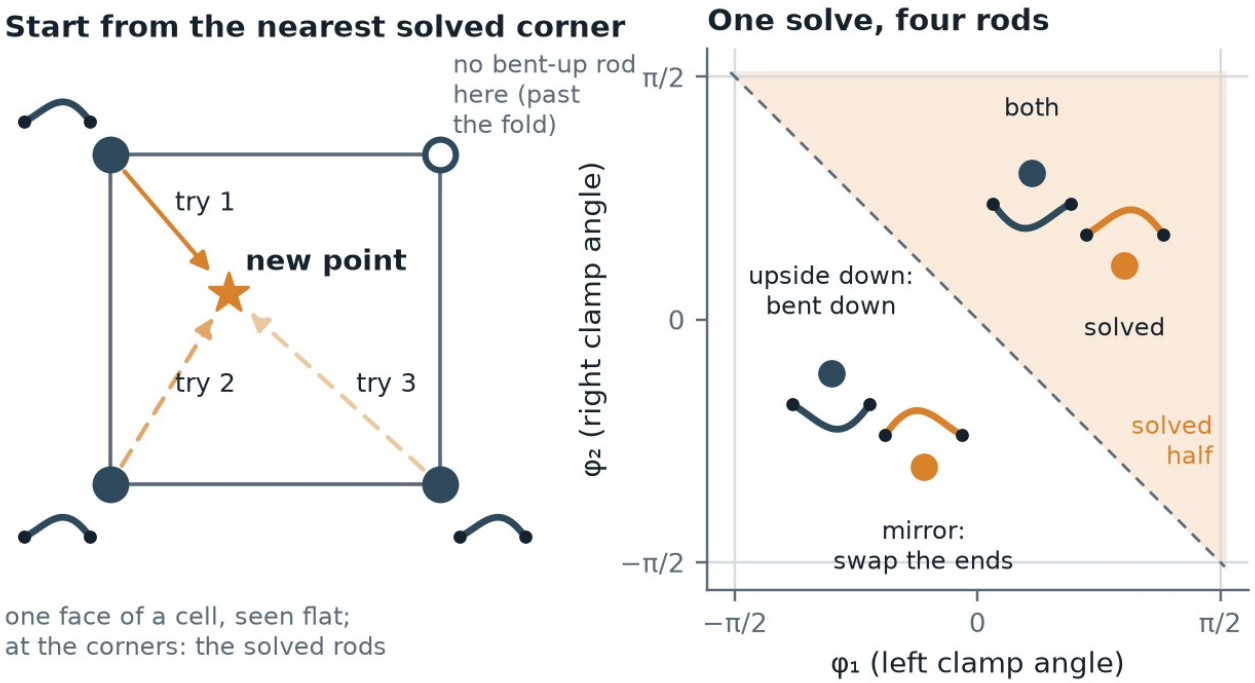
16 points in one cell, seen flat (φ_1 across, φ_2 up). Red: empty square; yellow: crowded square

1 **What a cell is**
The box between 8 neighbouring grid points, 0.01 wide in φ_1, φ_2 and d. Its corners are already solved.

2 **Spread evenly: Sobol**
A recipe that gives every part of the box its share of points: no clumps, no holes.

3 **A new shift in each cell**
Each cell slides its pattern by its own random amount, so neighbouring cells never share one pattern.

Solving each new point from its neighbours



998 k

rods accepted

from 1.11 M new points

112 k

“no rod here”

kept: past a fold

9

dropped

3 more: solved afresh

1 **Start from a nearby answer**

Each rod is found by Newton's method, which improves a guess step by step. A solved corner rod, at most 0.01 away, is a very good guess.

2 **Nearest corner first, up to 3 tries**

Only corners holding the bent-up rod are used. The answer is kept if it converged, did not snap to the other shape, and is stable.

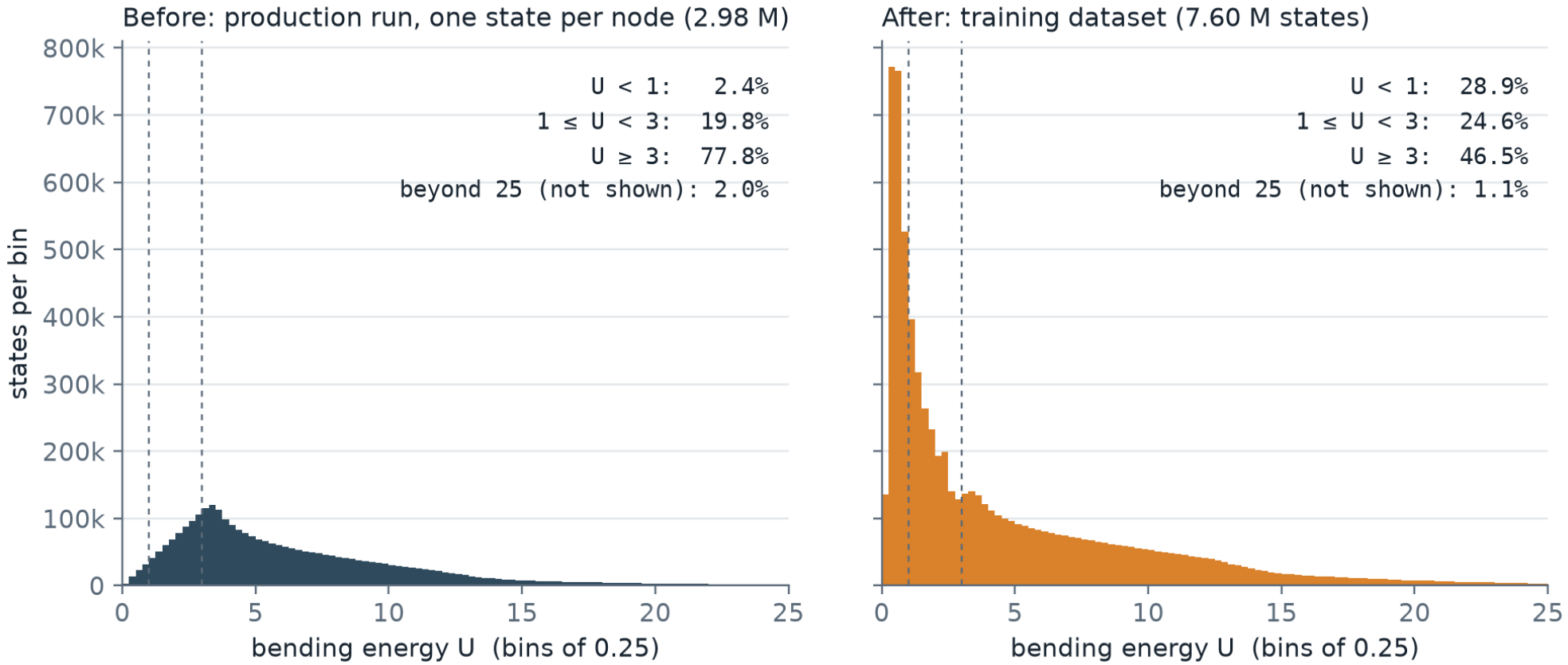
3 **If every try fails**

Near a fold the bent-up rod may not exist at that point: it is kept as a “no rod here” example, which shows where the shape ends. Elsewhere: a fresh solve, or the point is dropped.

4 **One solve gives four rods**

A rod seen from behind (ends swapped) or turned upside down is still a valid rod, so only a quarter of the points are solved.

Refinement moves the data to low energy

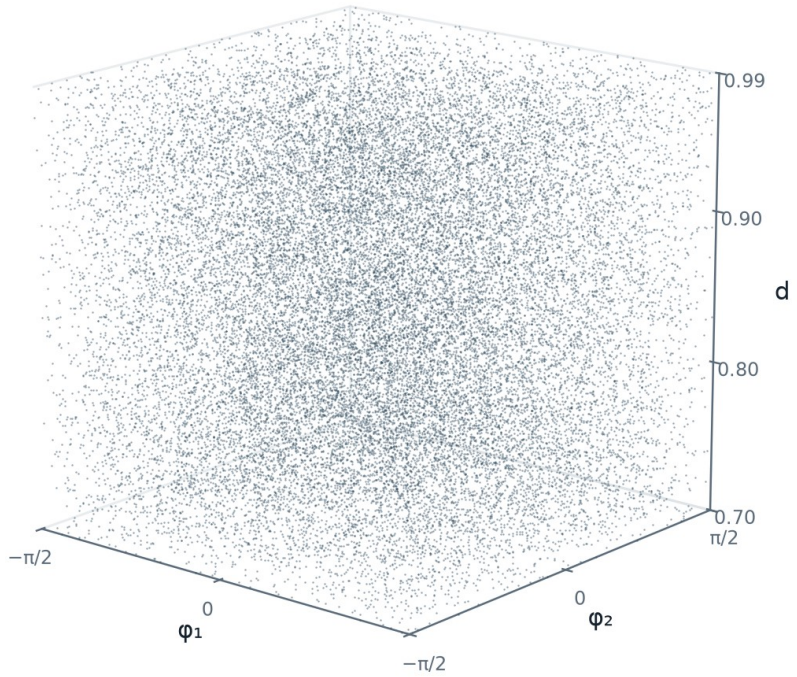


Same bins and axes. Before: one state per grid node, 2.4 % of them with U < 1. After: both branches at every node plus 4.0 M refined states (998 k solves and their symmetry images); 28.9 % with U < 1 and 24.6 % with 1 ≤ U < 3. The grid's high-energy states are all kept.

In 3D: the new points gather where the energy is low

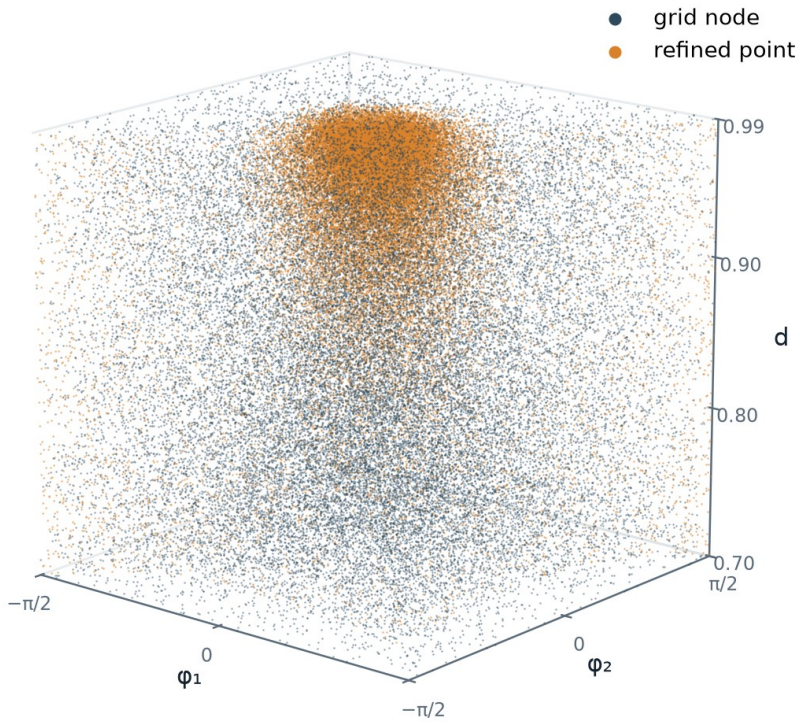
Before: 2.98 M grid states

one per grid node; 1 % drawn (29,897 dots)



After: 7.60 M states

both branches + refined points; 1 % drawn (76,088 dots)

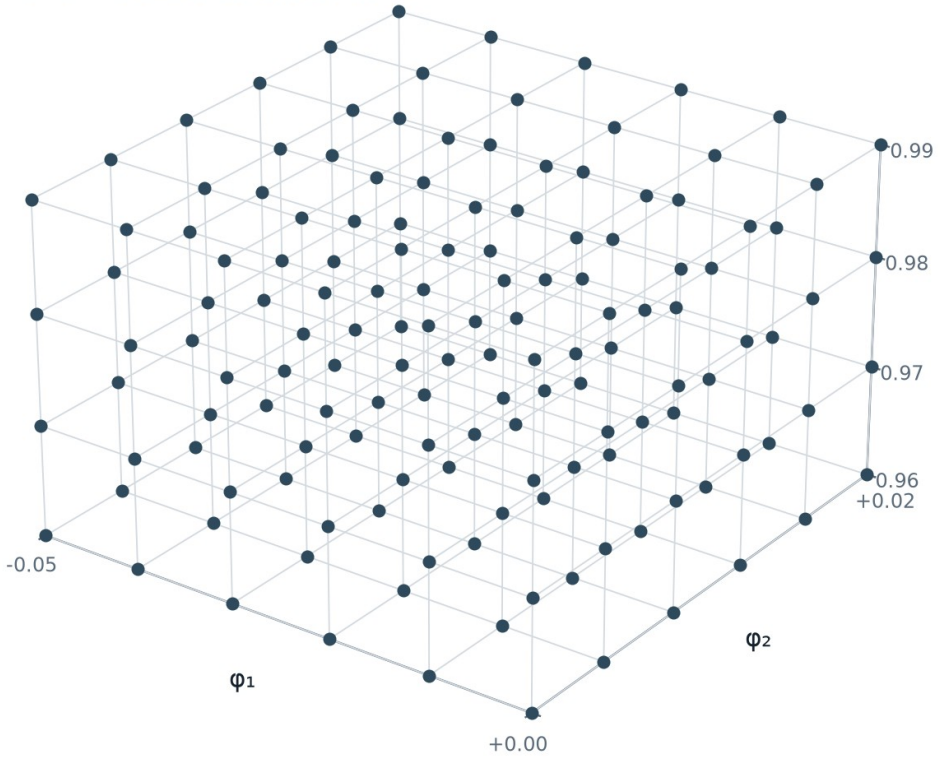


Each side draws the same random 1 % of its points, so dot density is data density. Before: one state per grid node, evenly spread. After: refined points (orange) fill the low-energy region at high d near the centre; grid nodes (slate) still cover the rest. Orange away from the centre comes from the minimum of 16 points per partial cell: fold edges, and the diagonal $\varphi_1 = \varphi_2$.

Close up: more points inside the cells where U is low

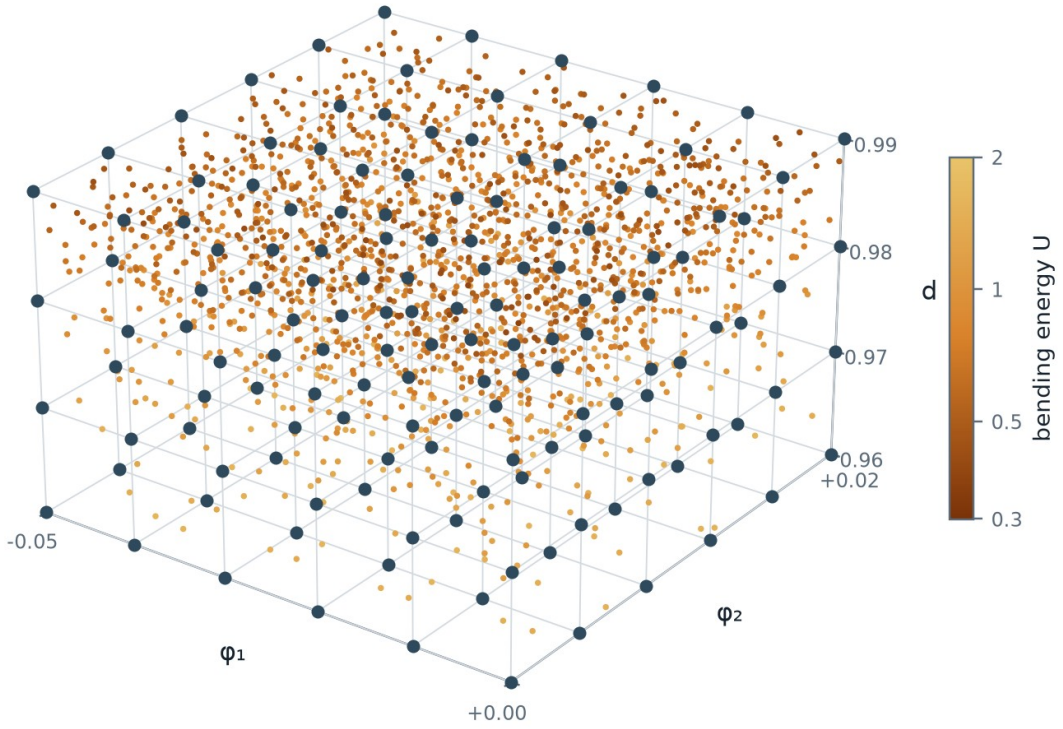
Before: 144 grid nodes

5 × 5 × 3 cells, a rod at every corner



After: 1,926 bent-up states

the same corners, plus points inside each cell



Every bent-up point in 75 cells at d = 0.96–0.99, $\varphi_1 \in [-0.05, 0]$, $\varphi_2 \in [-0.03, 0.02]$. The grid nodes stay; each cell adds its own points inside. Towards the top U falls (0.36 at the lowest) and the cells hold up to 64 points; at the bottom, where U reaches 1.8, as few as 5.